

Customer Churn Analysis using Tableau

PRESENTED BY:

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Agenda

Company Overview

Challenges and Objectives

Methodology

Data Cleaning & Joining

Analysis & Insights

Recommendations

Get in Touch

Company Overview

TeleWave Communications is a fictional telco company that provided home phone and Internet services to 7043 customers in California.

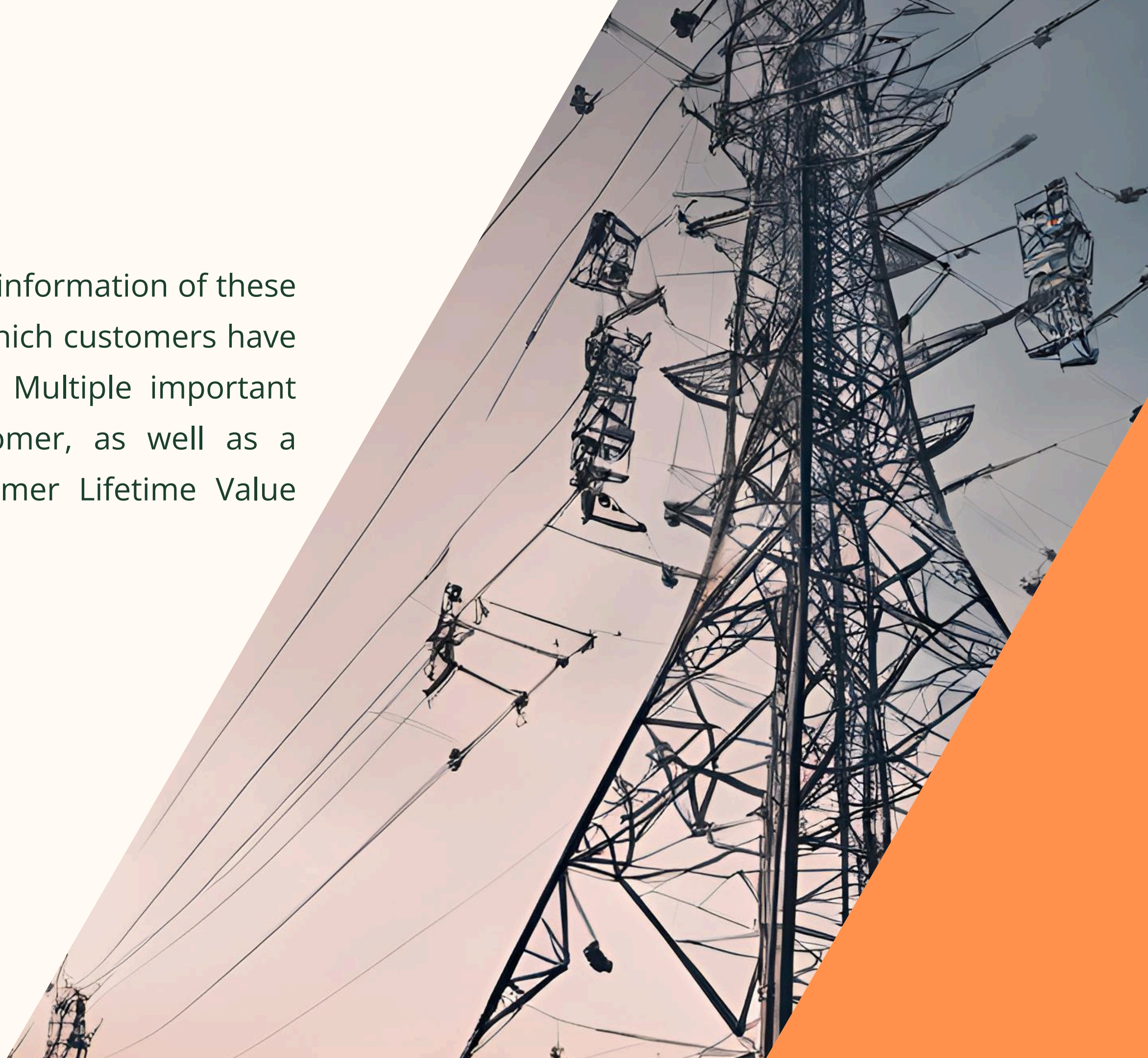
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Dataset

The dataset of customer churn data contains information of these 7043 customers in California in Q3 such as which customers have left, stayed, or signed up for their service. Multiple important demographics are included for each customer, as well as a Satisfaction Score, Churn Score, and Customer Lifetime Value (CLTV) index.

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Problem Statement

TeleWave Communications experienced a high churn rate of **26.54%** in **Q3** and seeks to **identify the underlying factors** contributing to this issue. A **comprehensive analysis** is required to uncover **key drivers of customer attrition**, provide **actionable insights** and **recommendations** to reduce churn and improve customer retention.

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Challenges and Objectives

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Challenges

- Identify underlying reasons for churn rate, build dashboard to present insights.
- Come up with recommendations to decrease this churn and improve customer retention.

Objectives

- Design an Exploratory Dashboard which summarises all key insights.
- Build individual visualisations to depict deeper insights.
- Recommendations to decrease overall churn rate.

Methodology

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01

Data Cleaning & Joining

- Perform data cleaning to remove errors and join all tables to get final dataset.

02

Analysis & Insights

- Showcase key insights observed during data analysis of cleaned data.

03

Recommendations

- Based on our insights, come up with recommendations to decrease churn rate.

01

Data Cleaning & Joining

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This step involves **removing errors, inconsistencies, missing data** and other steps to ensure clean data for data analysis.

We will using **MS Excel to perform data cleaning** and then join our data in **SQL** to obtain final cleaned dataset for further analysis.



Data Cleaning

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- 1** Imputed missing values in ``Churn Reason`` & ``Churn Category`` column of ``Status`` table with ``Not Applicable`` as missing values were only in the customers who did not churn.
- 2** Ensured data to be complete (with no missing values), accurate (looked for inconsistencies and removed them if available) and removed unnecessary columns which do not add any value to table.
- 3** Finally, merged all the tables as left joined all tables as (on common field "**Customer ID**"):
`services - - - > status - - -> demographics - - -> location - - -> population`

02

Analysis & Insights

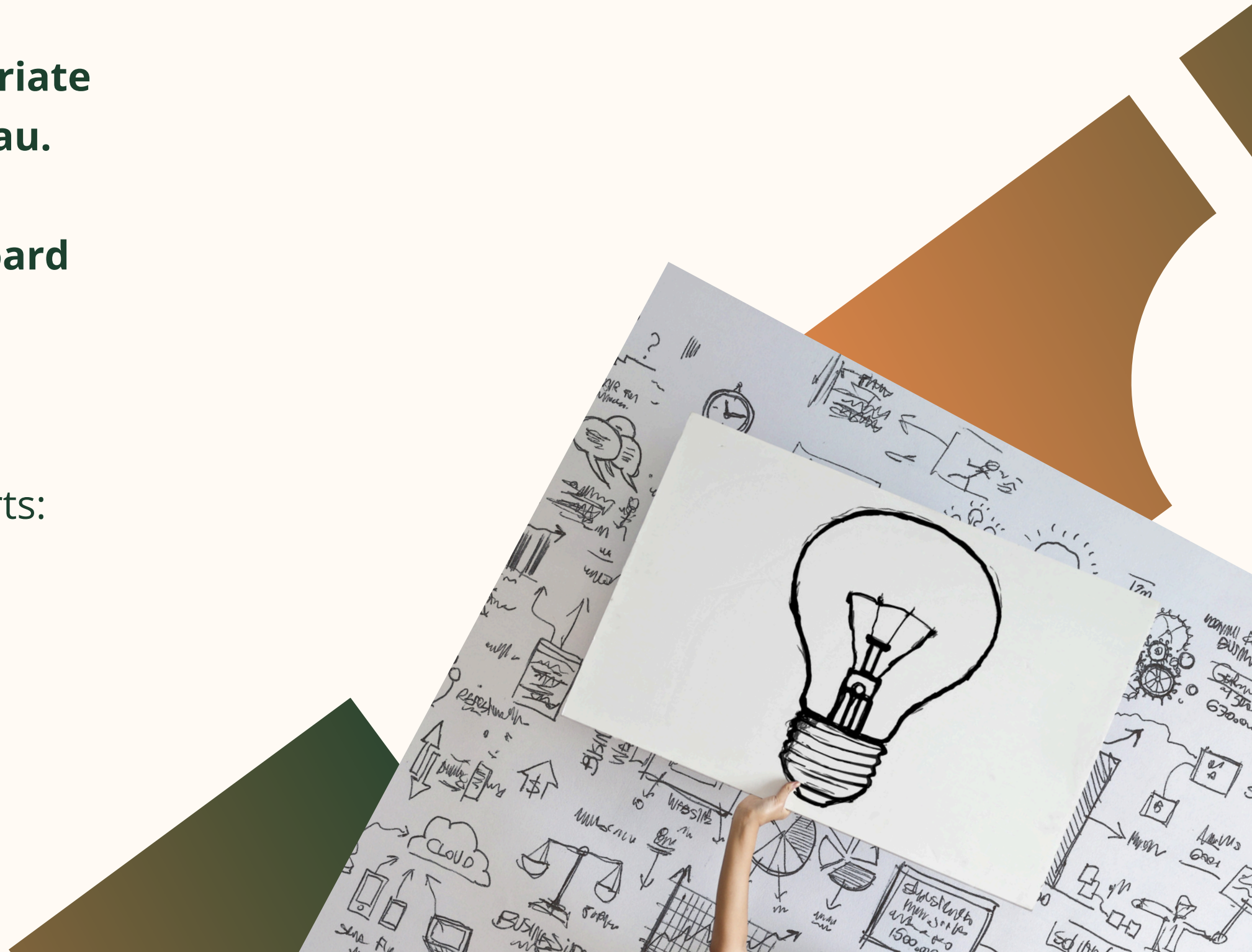
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This step involves **Univariate, Bivariate and Multivariate analysis of variables** and **drawing insights in Tableau**.

Apart from building visualisations, **a dynamic dashboard would also be added.**

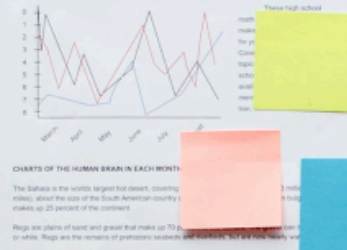
This analysis is further divided into following three parts:

- **Overview**
- **Customer Analysis**
- **Service Analysis**

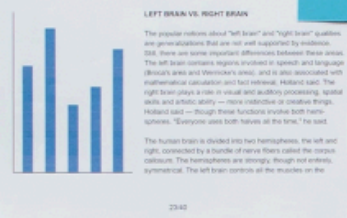


We are creating a system where very talented creatives can develop their own style and share everything. Some young talented kid comes along who can take better images than me. I'll open her images and add cherry to it to regain the number one spot. An amazing photographer takes a photo in New York and uploads it. By the end of the day, zero amazing researchers have already added their graphic concepts to it. We keep hiring the most talented young people, we are doing this right now here in Thailand and soon we'll be able to start attracting the best young creatives from around the world. We don't need people with tons of experience, just brilliant artists straight out of uni or not.

Then there is collaboration. We can hire creatives for specific projects. Hire a famous photographer for a shoot. Contract a font foundry to create a set of fonts for us.



The Gattuso is the world's largest hot desert, covering more than 500,000 square miles (1,300,000 square kilometers). It is located in the Sahara Desert in North Africa. The Gattuso is the world's largest hot desert, covering more than 500,000 square miles (1,300,000 square kilometers). It is located in the Sahara Desert in North Africa.



LEFT BRAIN VS. RIGHT BRAIN

The popular notion about "left brain" and "right brain" qualities are generalizations that are not well supported by evidence. Still, there are some important differences between these areas. The left brain contains regions involved in speech and language (Broca's area and Wernicke's area), and is also associated with mathematical calculation and fact retrieval. Holstad said: "The right brain plays a role in visual and auditory processing, spatial skills and artistic ability — more intuitive or creative things. Holstad said — through these functions involve both hemispheres. "Everyone uses both halves of the brain," he said.

RIGHT BRAIN VS. LEFT BRAIN

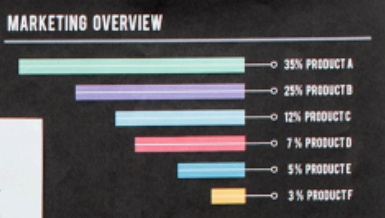
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Actually, all stock photographers work alone or in small teams. They are all competing with each other. Nobody has ever really managed to scale. Several people have cloned themselves to produce more of the same look.

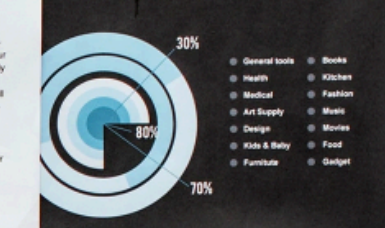
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Although we are in the micro stock category we aim to produce better than the images in the macro stock category. We are in a position to invest more in shoots than almost all of our competitors.



Company has been dedicated to creating fresh and amazing imagery for decades. We do better together. You need visual elements that reflect your world, concept, styling, and professional quality. The result? Endless potential applications. We put a lot into our photos. Company has been dedicated to creating fresh and amazing imagery for nearly a decade. We're passion-driven, not profit-driven. Our future: Tell us what you need from a stock photo — we're dedicated to creating fresh and amazing imagery for nearly a decade.



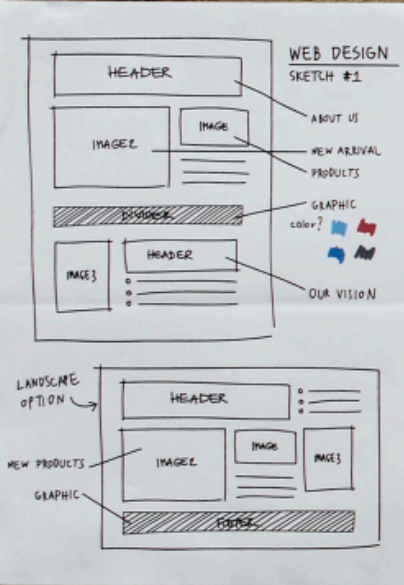
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reflect your world and your brand. Our images combine quality. The result? Authentic, balanced work with endless potential applications. We put a lot into our photos, so you can get a lot out of them. Company has been dedicated to creating fresh and amazing imagery for nearly a decade.

driven. Our photos, like us, are real and genuine. Tell us what you need from a stock photo — we're dedicated to creating fresh and amazing imagery for nearly a decade. We do better together. You need visual elements that reflect your world, concept, styling, and professional quality. The result? Endless potential applications. We put a lot into our photos, so you can get a lot out of them. Company has been dedicated to creating fresh and amazing imagery for nearly a decade.

OVERVIEW



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PLANNER & CALENDAR

	SUNDAY	MONDAY	TUESDAY	WEDNESDAY	THURSDAY	FRIDAY	SATURDAY
TO DO				1	2	3	4
TO DO				5	6	7	8
TO DO				9	10	11	12
TO DO				13	14	15	16
TO DO				17	18	19	20
TO DO				21	22	23	24
TO DO				25	26	27	28
TO DO				29	30	31	



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June 23rd

- Meeting
- Workshop
- Development
- Meet sponsors

LEFT BRAIN VS. RIGHT BRAIN

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Overall Customers & Churn Rate

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High churn rate of **26.54%** was observed during **third quarter** overall.

- Out of **7043 total customers**, **1869** (26.54%) have stopped using services offered by TeleWave and are **not customers anymore**.

# of Customers	# of Churned Customers	Churn Rate
7,043	1,869	26.54%

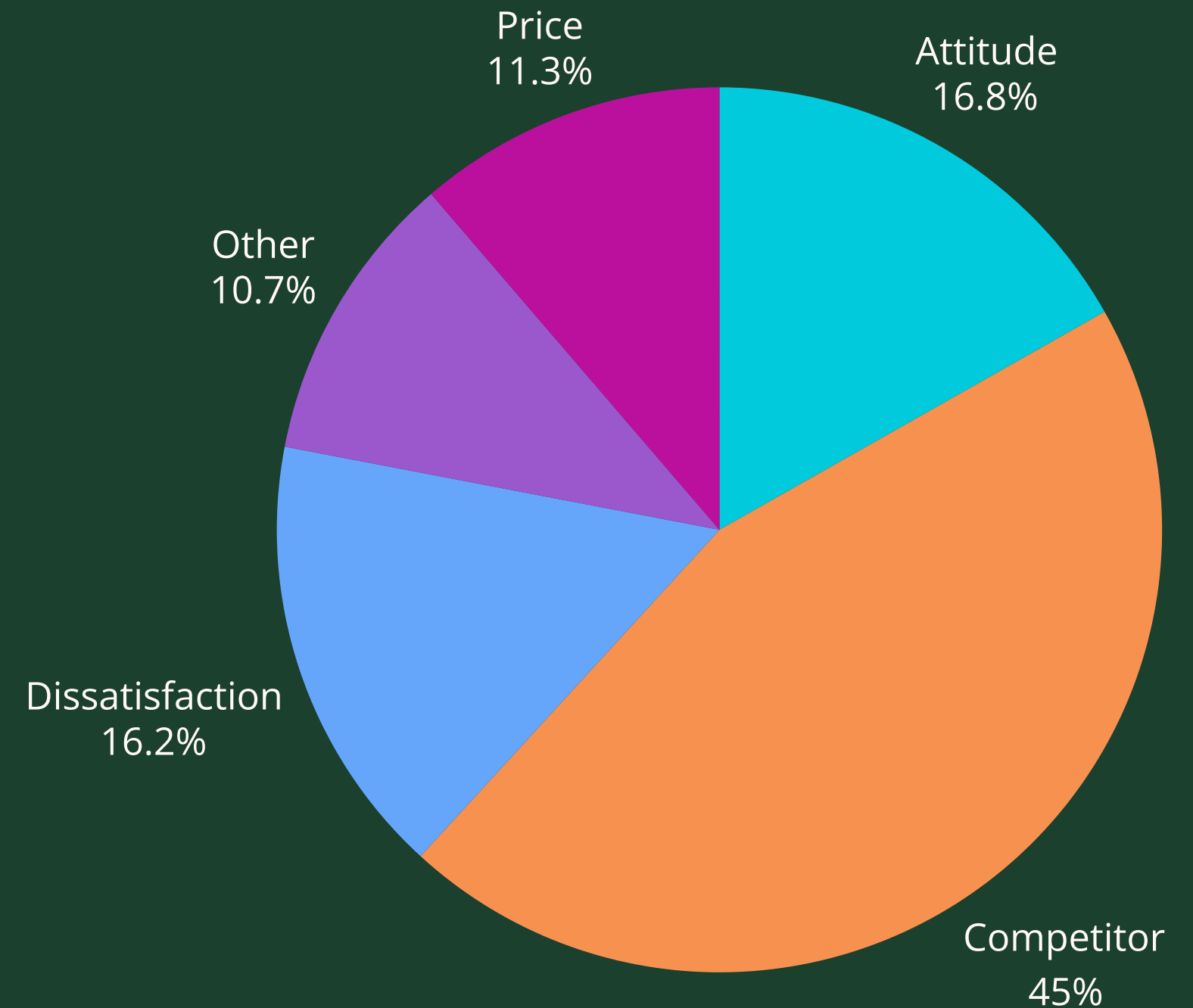
Customers and Churn Rate

Churn Category

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Competitor is a major reason for nearly half of churned customers.

- **Competitor** (45%), **Attitude** (17%) and **Dissatisfaction** (16%) are the most popular categories for customer churning.



Churn by category

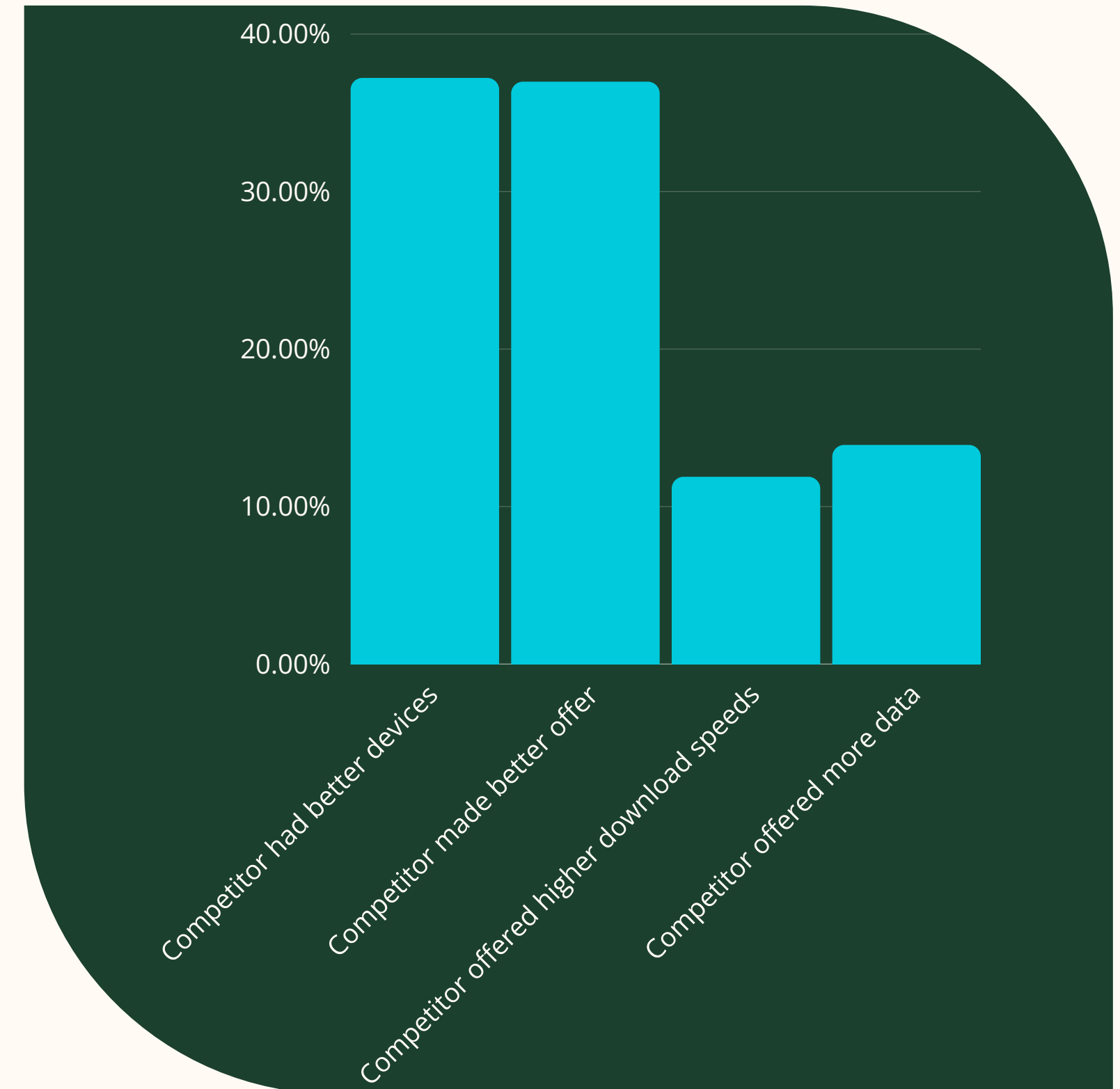
Let us look deeper into each category to understand better...

Competitor (Subcategory)

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Competitor had **better devices** and made **better offers**

- Competitor is offering better devices, better offers, more data, and higher download speeds

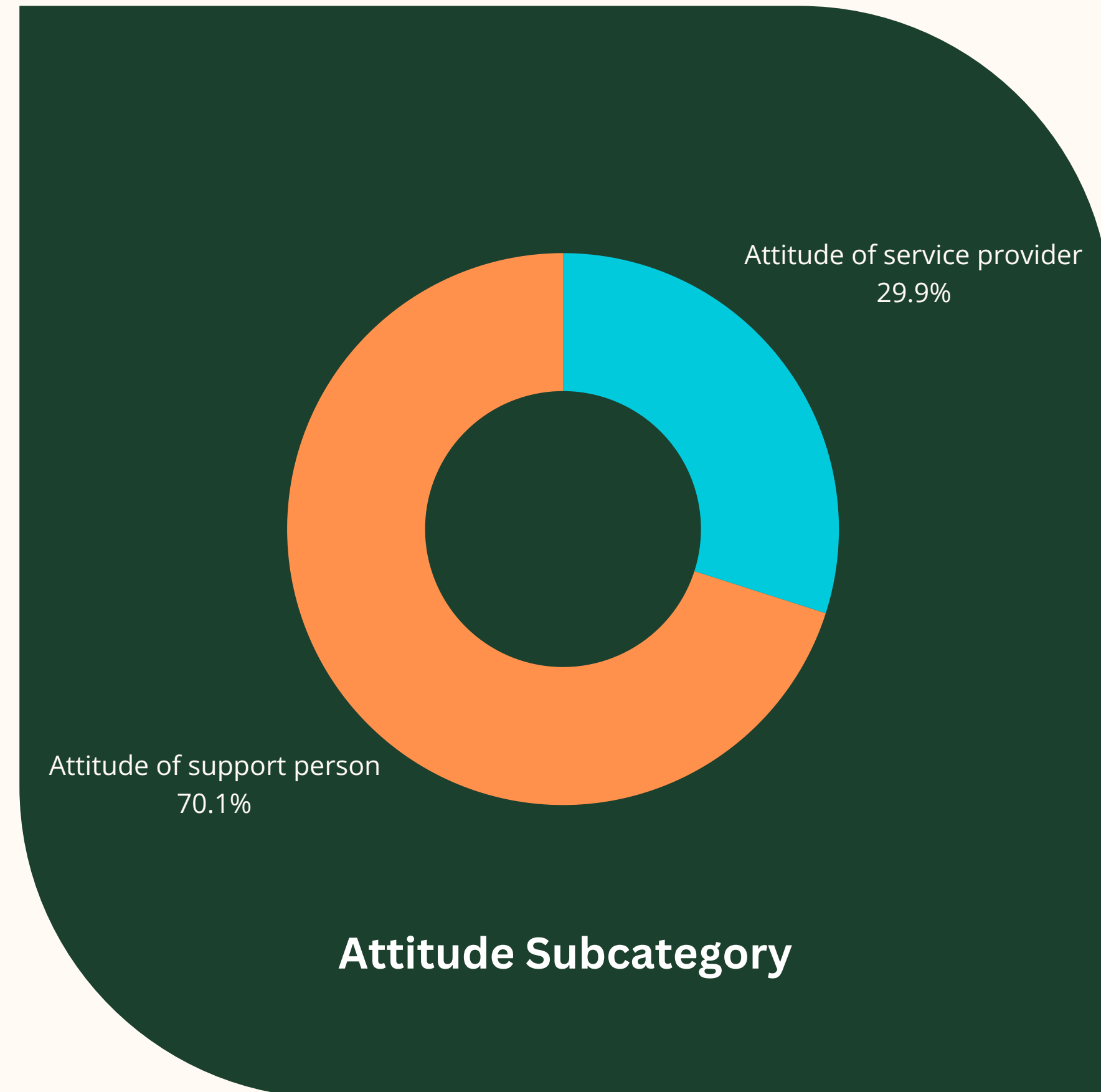


Attitude (Subcategory)

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Attitude of support person is a major concern.

- Roughly **17%** of churned customers have **attitude** as reason to churn.
- Digging deeper **support person (70%)** and **service provider (30%)** attitudes are leading to more customers dropping.

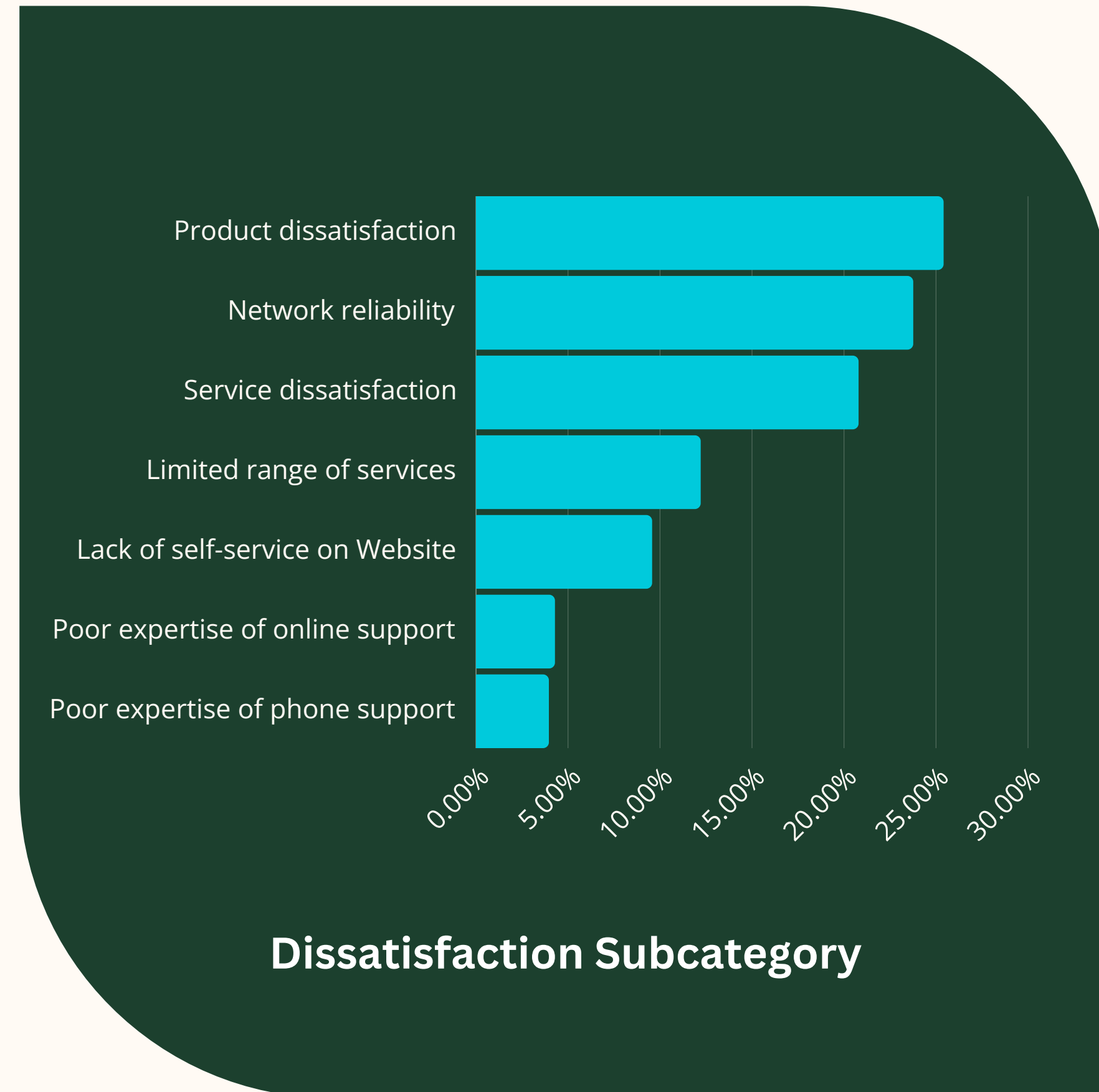


Dissatisfaction (Subcategory)

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Product, Network Reliability and **Service Dissatisfaction** are big **reasons** for customers getting dissatisfied and **churning**.

- Dissatisfaction with basic facilities as well as limited range of devices, poor website interface and poor customer support are leading to overall dissatisfaction amongst customers.

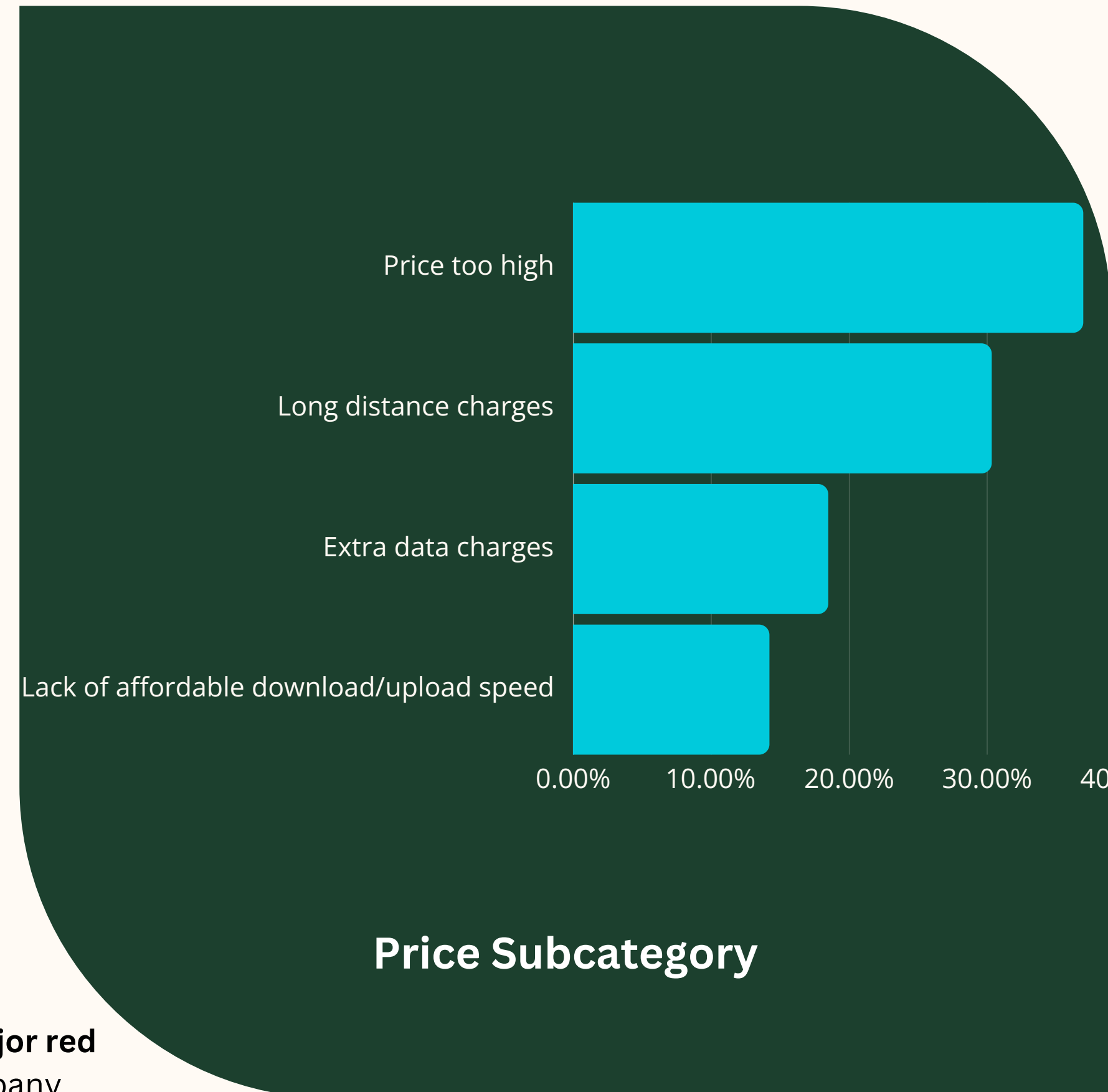


Price (Subcategory)

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Prices being too high, long distance charges along with extra data charges are also reason for customers churn.

- Not only the customers who churned dissatisfied with the services, other churned customers are getting overcharged with compromised quality.



From the **company's end** in terms of services, **Attitude and Dissatisfaction** are **major red flags for churning**. These are service issues and can be worked upon by the company.

Customer Analysis

In this step, we are going to **look into our customer and their overall demography**. This would help us in **understanding our customer better, identify churn trends** and **develop strategies to decrease** it.

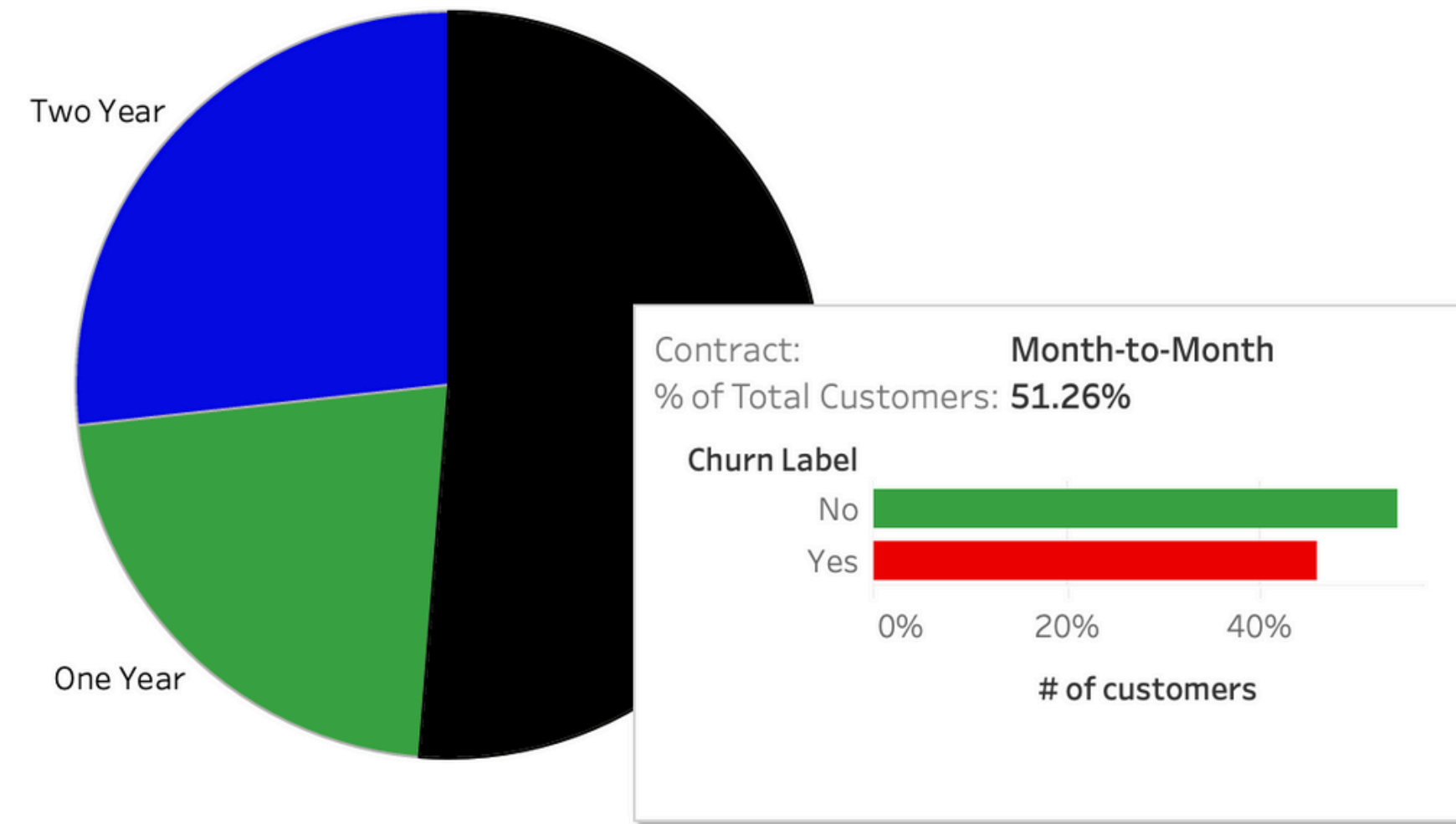


Churn by Contract

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Month-to-Month contract customers with **46%** churn rate

- **Month-to-Month** customers comprise **51%** of the customers overall.
- In Contract, **One Year** and **Two Year** contract customers have **significantly lower churn rate (11% and 3% respectively)**



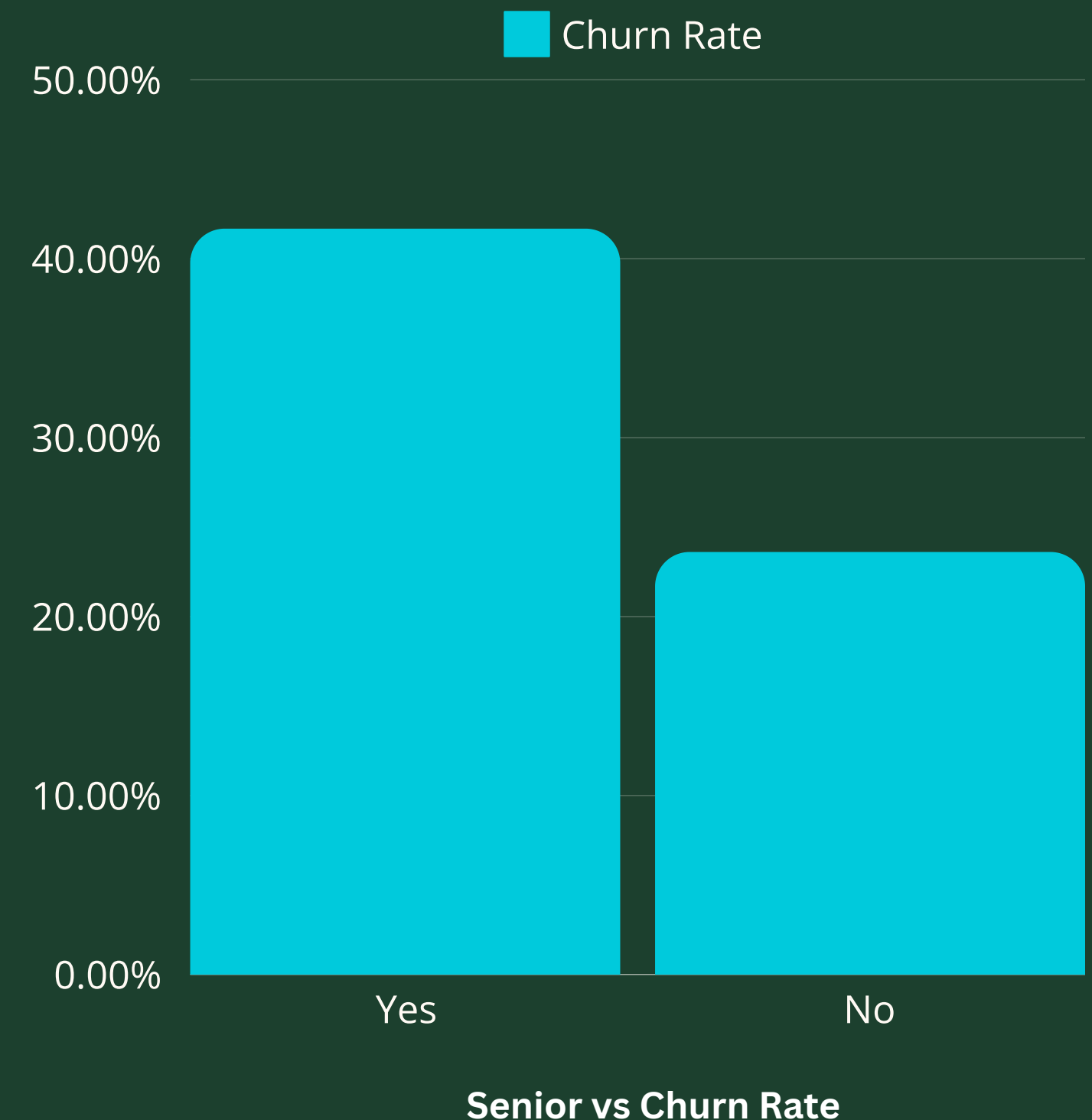
Month-to-Month churn

Senior vs Churn

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Senior customers are **twice** as likely to **churn**.

- Customers over 65 (**Senior**) have nearly double the churn rate than the ones who are below 65 (**Senior = 42%, Non Senior = 24%**)

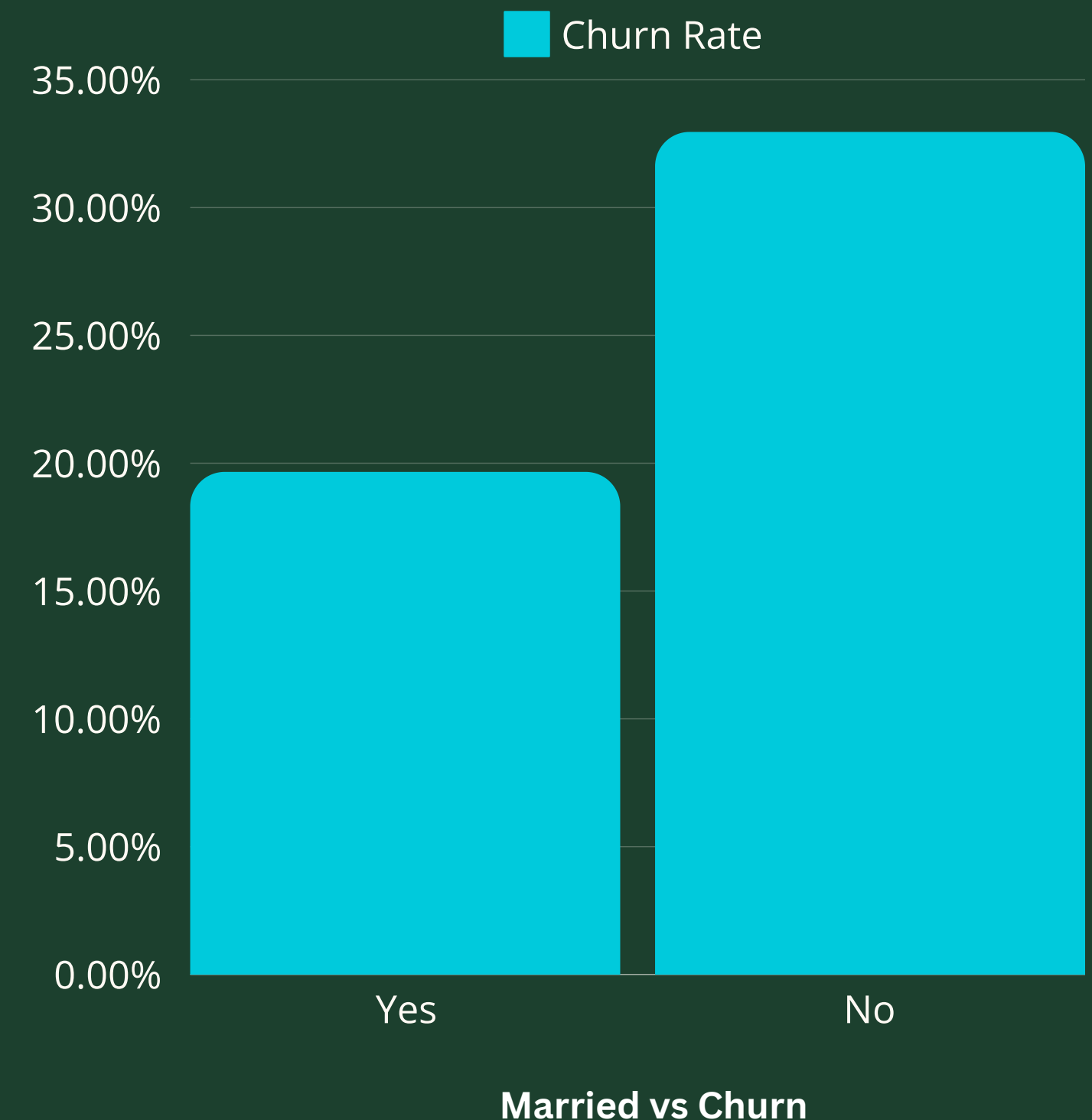


Married vs Churn

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Married customers are less likely to churn.

- **Married customers** have **significantly lower churn rate** (nearly **20%**) than **unmarried customers** (roughly **33%**).

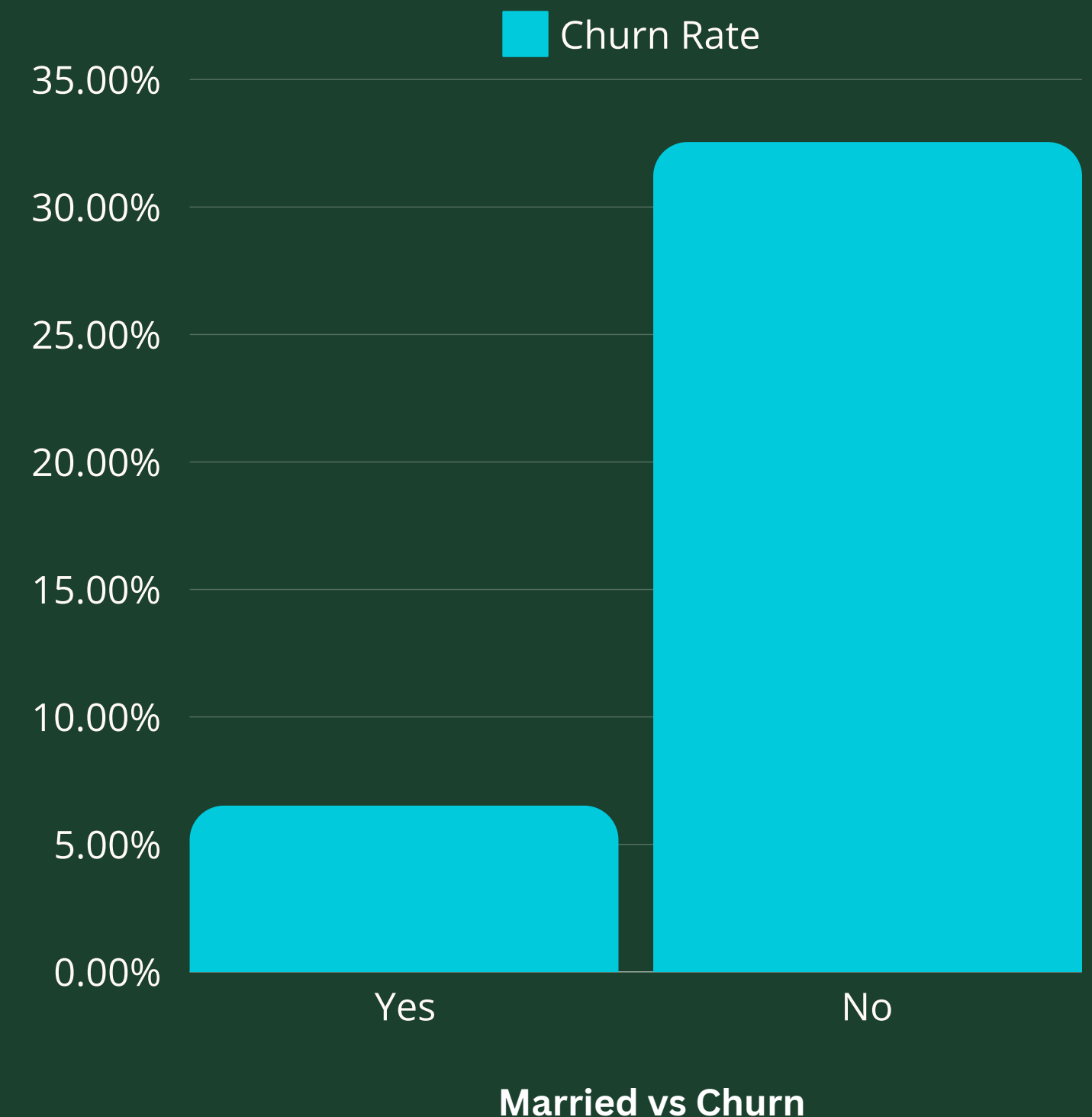


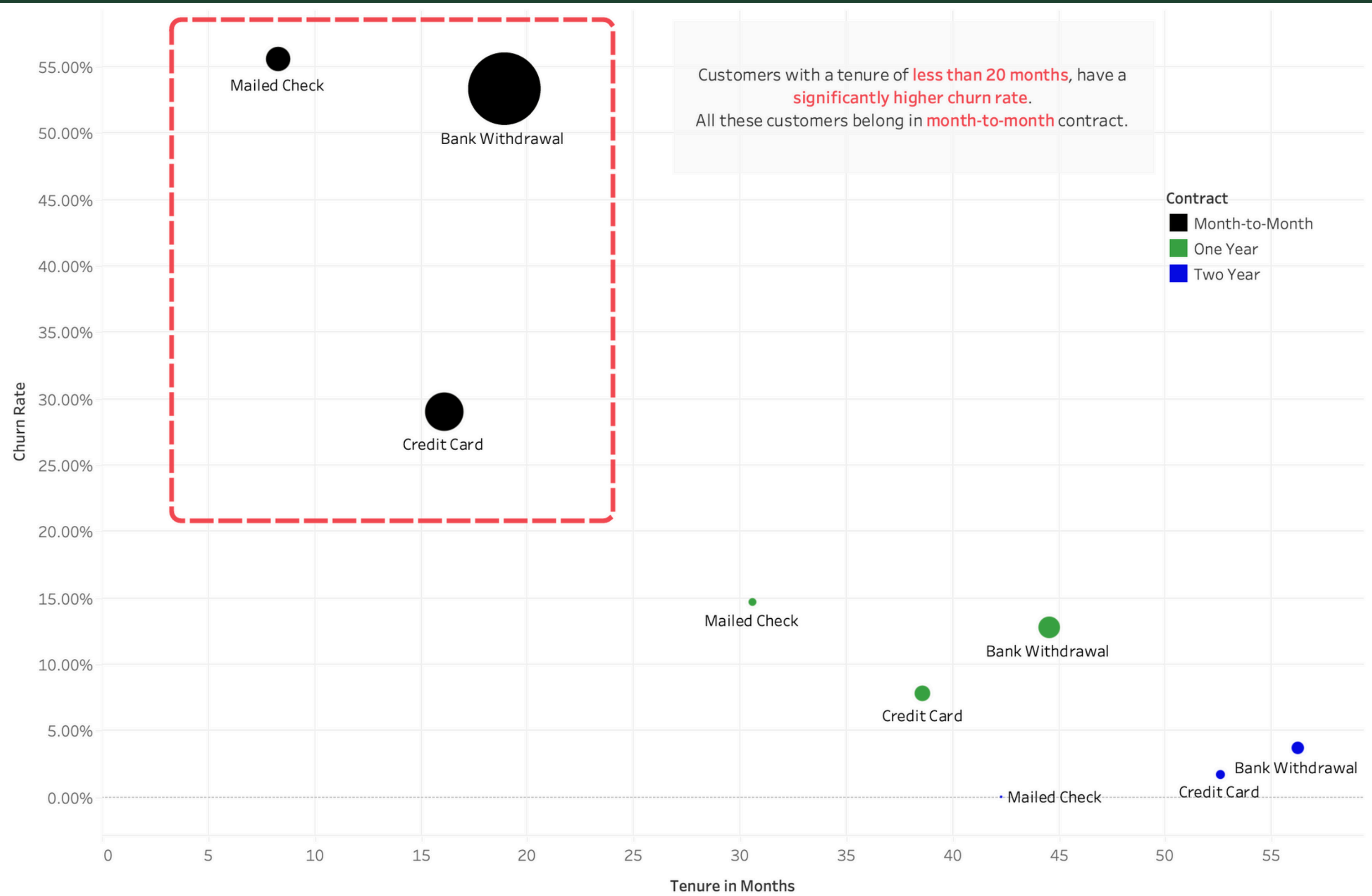
Dependents vs Churn

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Customers with **no dependents** are **four times more likely to churn**

- **Customers with dependents** have **significantly lower churn rate** (nearly **7%**) than **independent customers** (roughly **33%**).





Tenure, Payment Method, Contract vs Churn Rate

Insights?

Recent Customers with Shorter Tenure Churn Due to Competitors and Poor Support, While Long-Term Users Churn Less

- **Customers with Average Tenure** (in months) nearly **20 months** (or roughly two years) which means these are customers which started using company services only recently, **have churned**. And these are **all month-to-month contract users**.
- Their major reason is **competitor offering better devices** and better offers (external reasons) and **attitude of support person** (internal reason) as a major **reason to churn**.
- There is a **significantly lower churn rate** in customers with **average tenure over 20 months (churn rate nearly halves)** and their **churn reason** is either **Competitor or Product Dissatisfaction**.



Service Analysis

In this step, we are going to **look deeper into services offered by TeleWave Communications and its effect on overall churn**. This would help us in **identifying faulty services and issues from company's end** which are leading to customer churn.

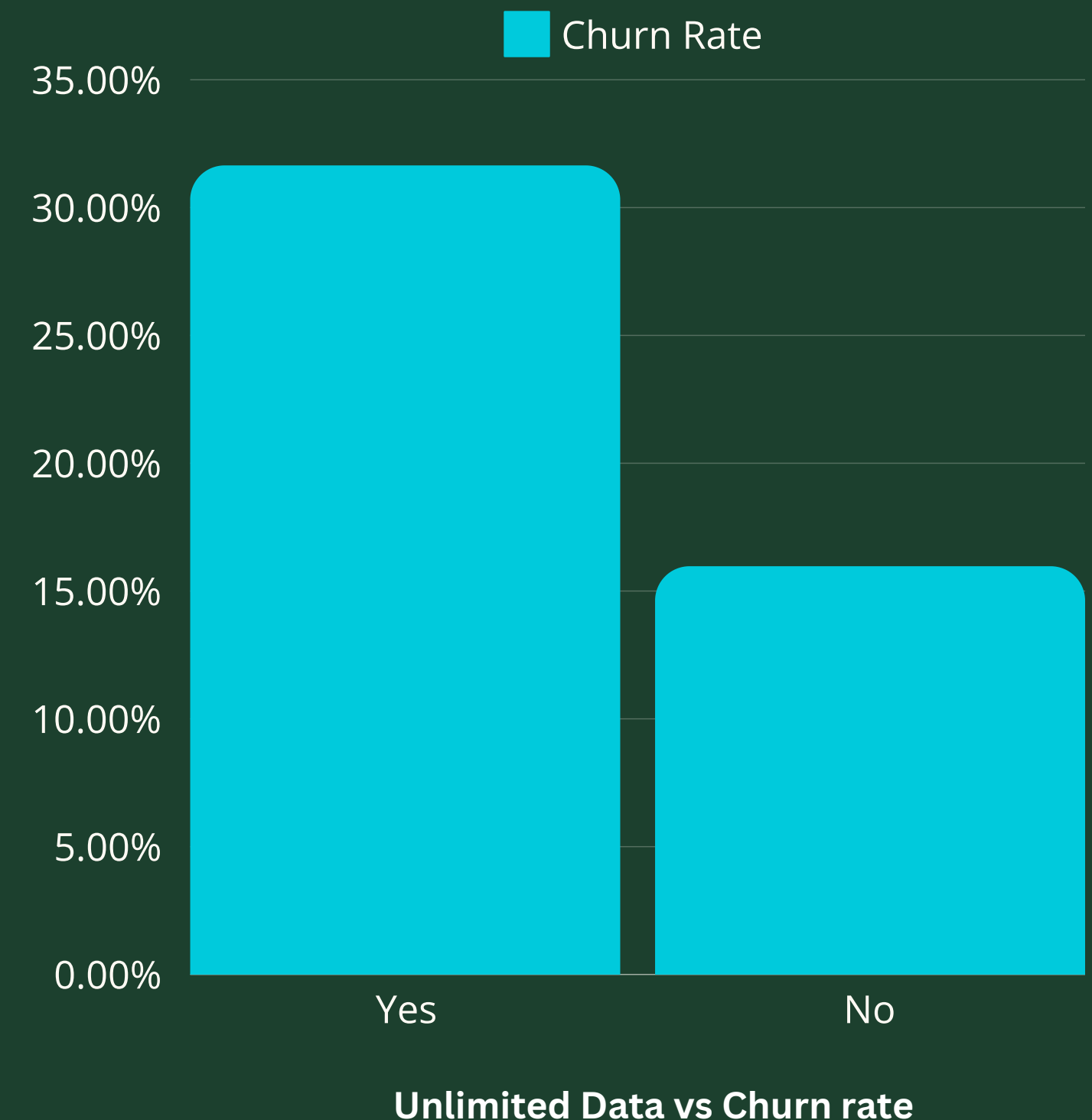


Unlimited Data

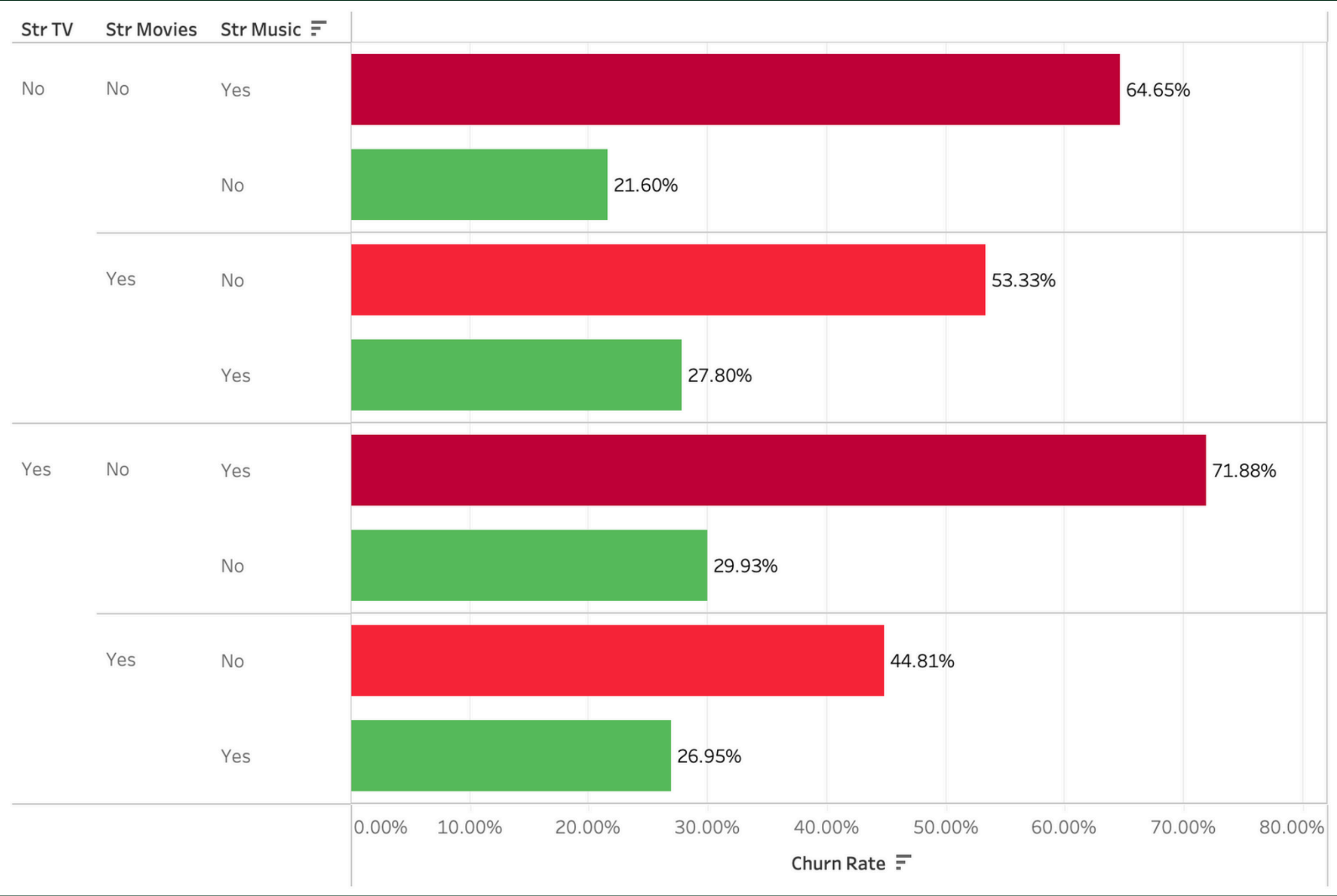
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Customers with **unlimited data** are **twice as likely to churn** compared to ones with **no unlimited data**

- Customers with **Unlimited Data** have a churn rate **32%** compared to ones with **no unlimited data** **16%**



Streaming Services vs Churn



It can be clearly seen that there are many red markers indicating high churn rates

Streaming Services with Churn Rates

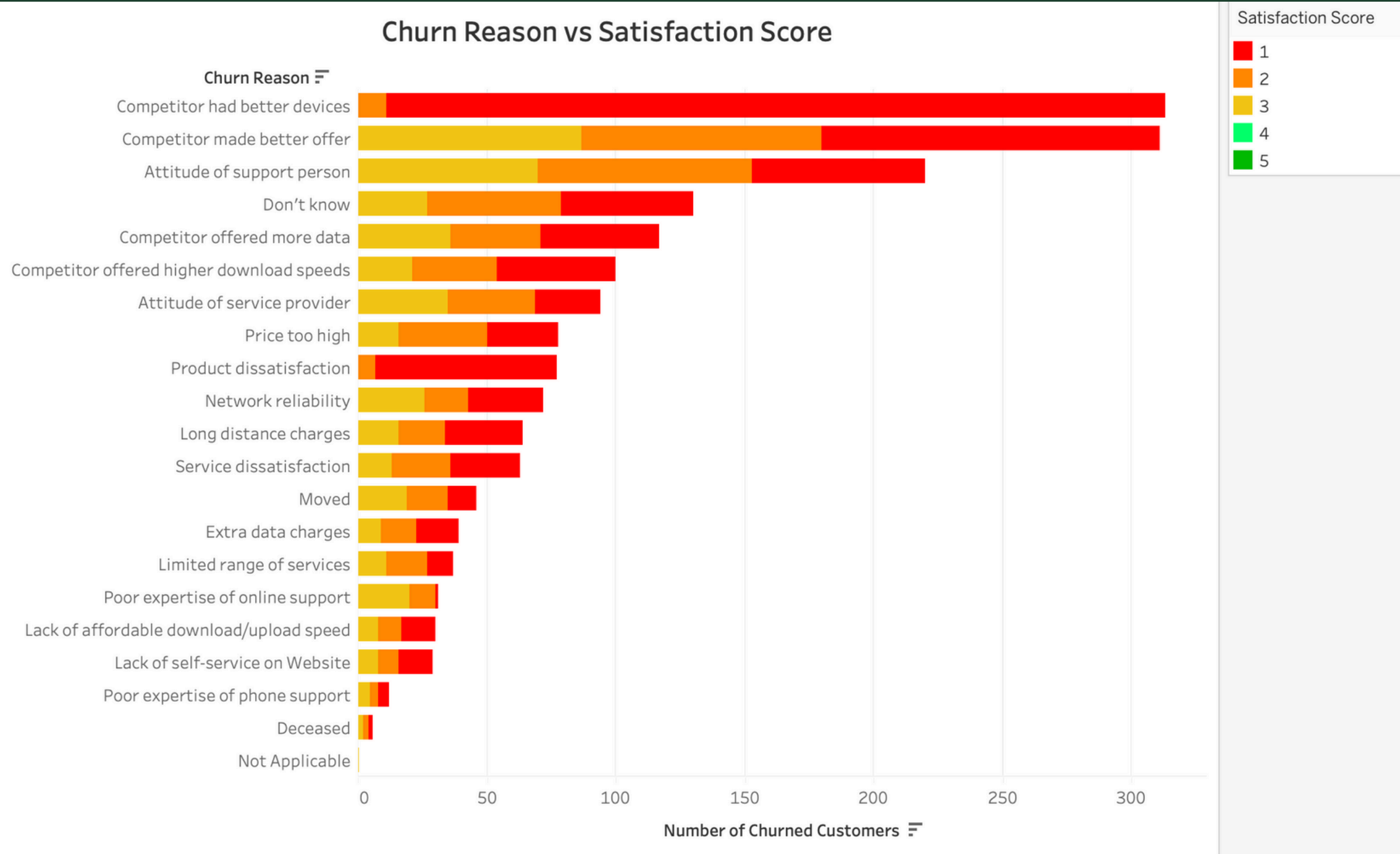
Insights?

TeleWave has **very poor music and movie streaming services**

- **Regardless of** customer having using a **streaming TV (Str TV)**, if they use internet service for **streaming music**, their **churn rate is very high (more than 60%)**. If they use the services for **streaming movies**, then too their **churn rate is very high (more than 40%)**
- Similar is true for streaming only **Music customers (churn rate is above 60%)**



Churn Reason vs Satisfaction Score



Customers have given very low satisfaction scores overall

Churn Reason & Satisfaction Scores

Insights?

TeleWave's devices (such as modems) are problematic and have been giving **poor experience** to the customers

- Of the **customers that left**, their **Satisfaction Scores** are all **1 or 2** for **Churn Reason = Competitor had better devices and Product dissatisfaction**.

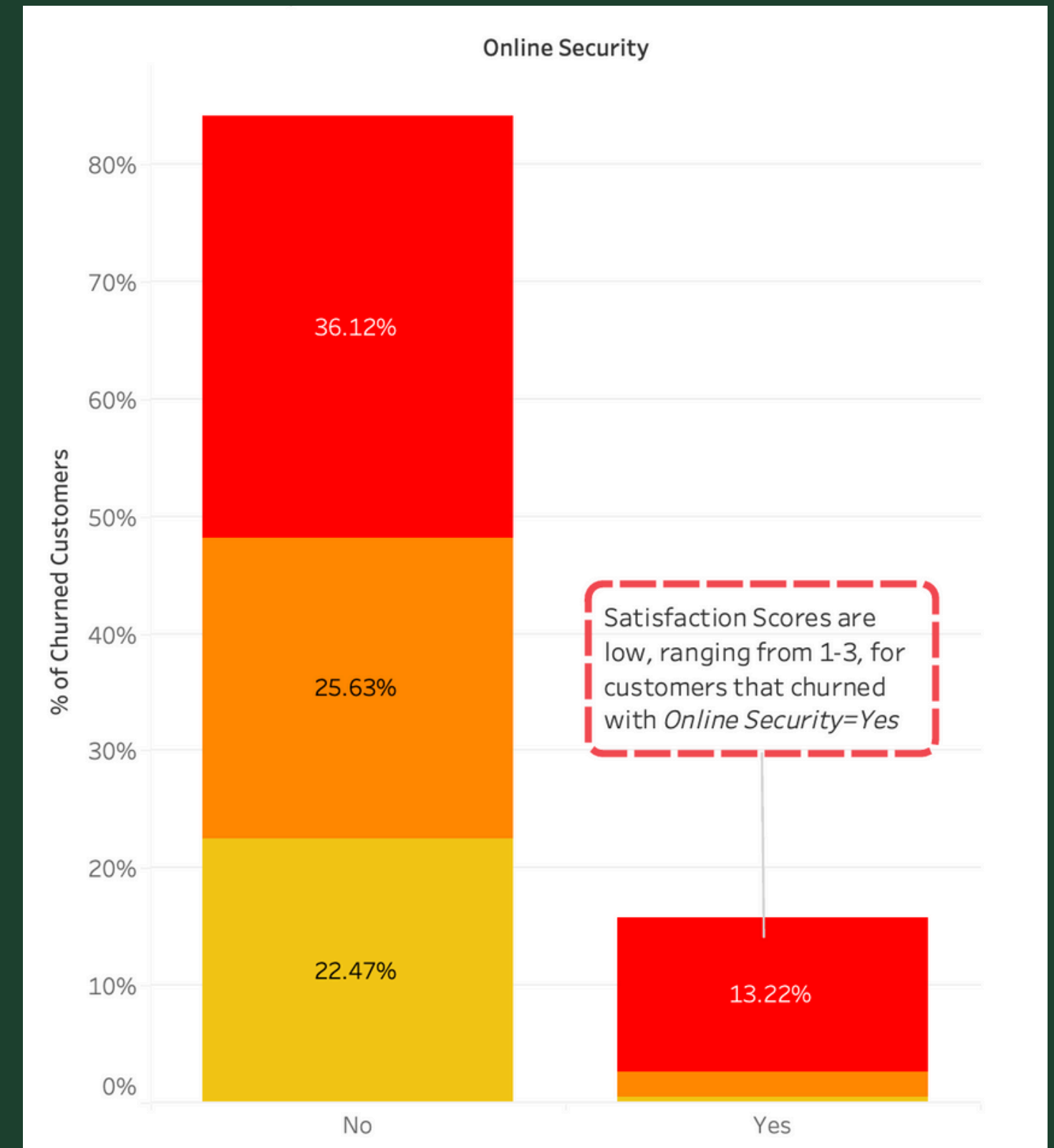


Online Security Services

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Customers give very **low satisfaction score** to company's **online security services**

- Customers **with Online Security** feature have given **low satisfaction score**, i.e., 1, clearly indicating that the company's **online security services is bad**.



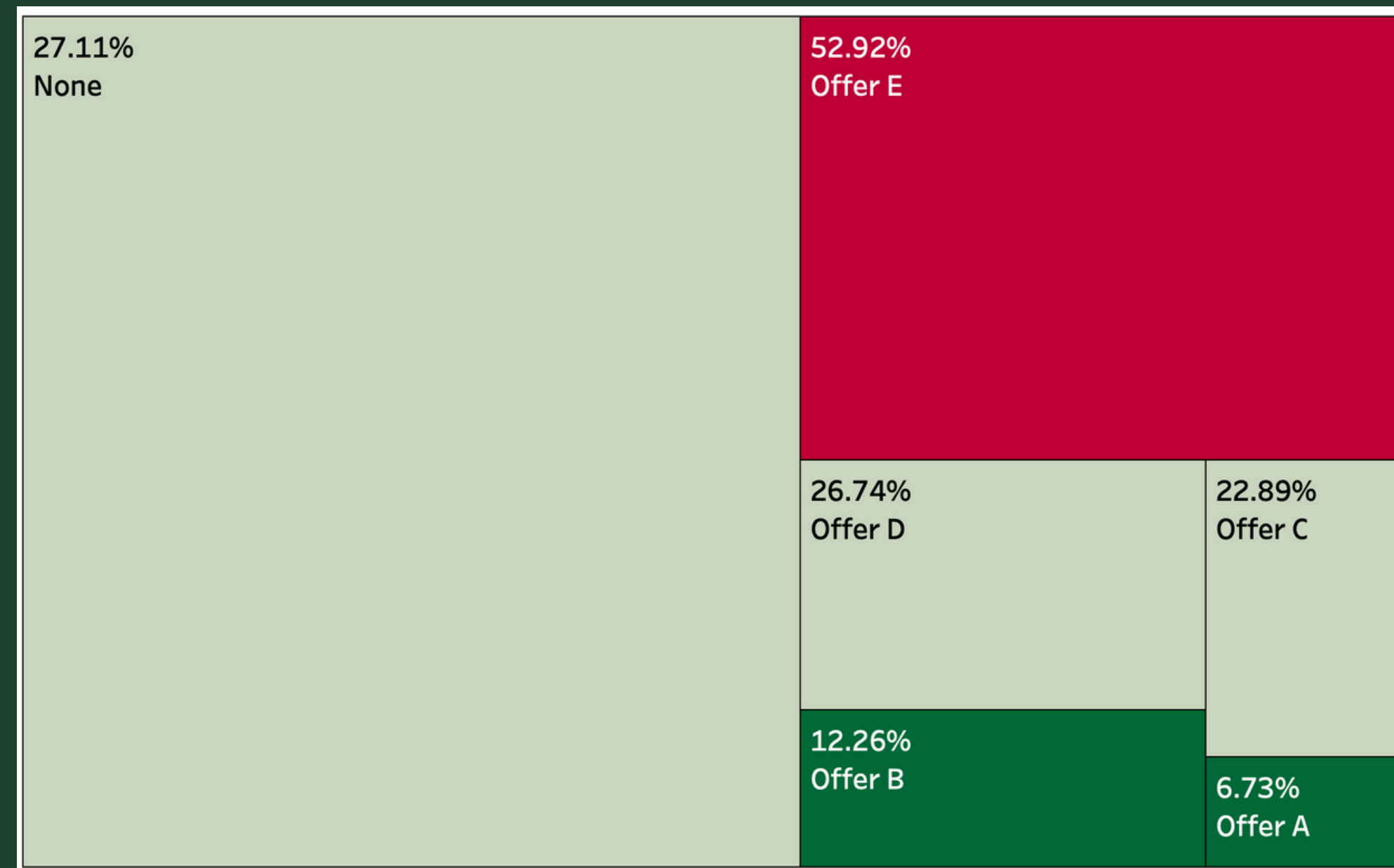
Online Security vs Satisfaction

Offer vs Churn Rate

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Customers with **offer A** have **lowest churn rate**

- Size of the quadrant depicts total number of churned customers.
- Customers with offer E have highest churn rate (~53%).
- Customers with offer A have lowest churn rate (~7%)



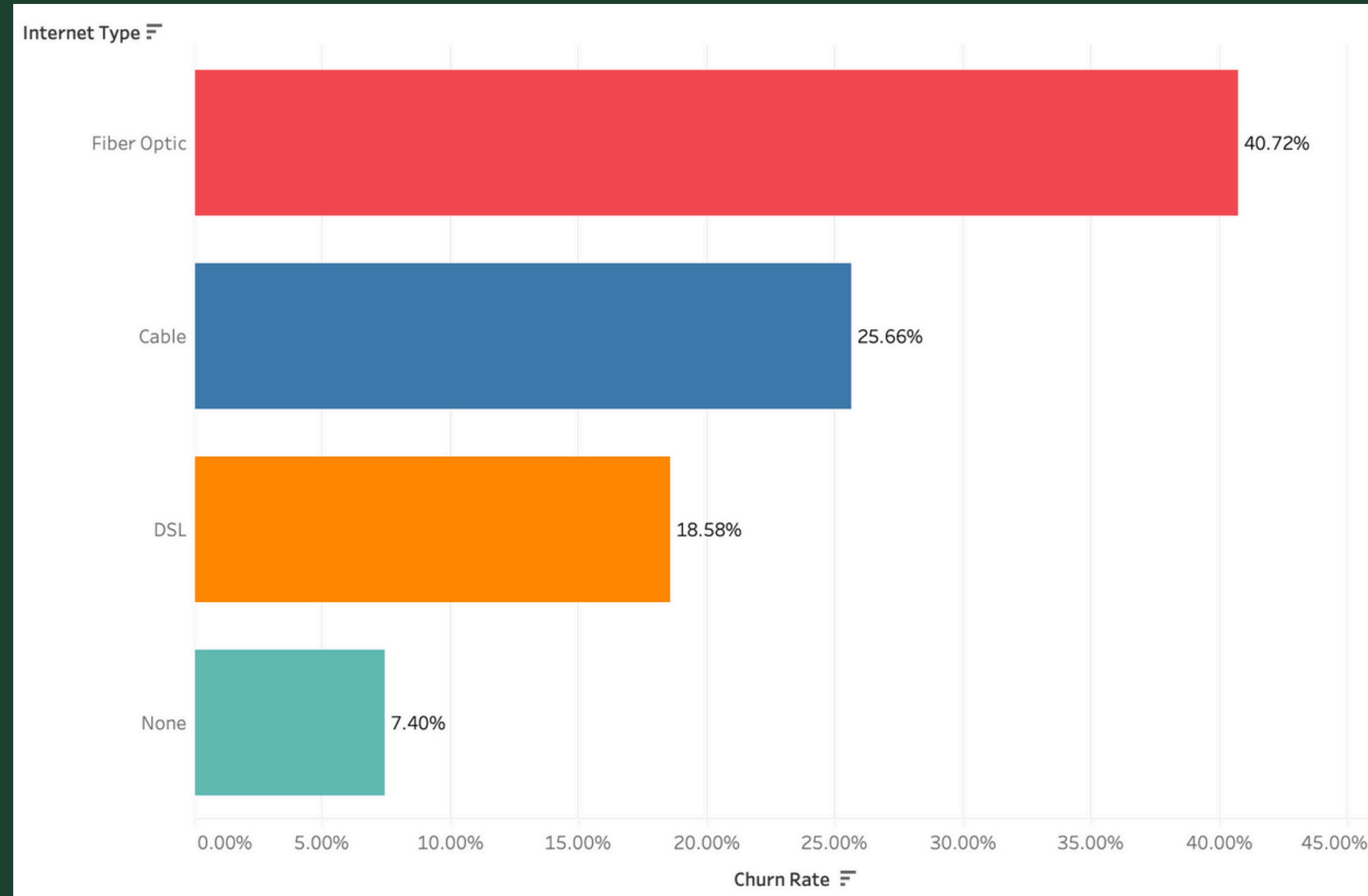
Offers vs Churn Rate

Internet Type vs Churn

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Fiber Optic has highest churn rate

- Majority of **churned customers (nearly 40%)** had fiber optic as their internet type followed by **Cable (26%)** and **DSL (19%)**.



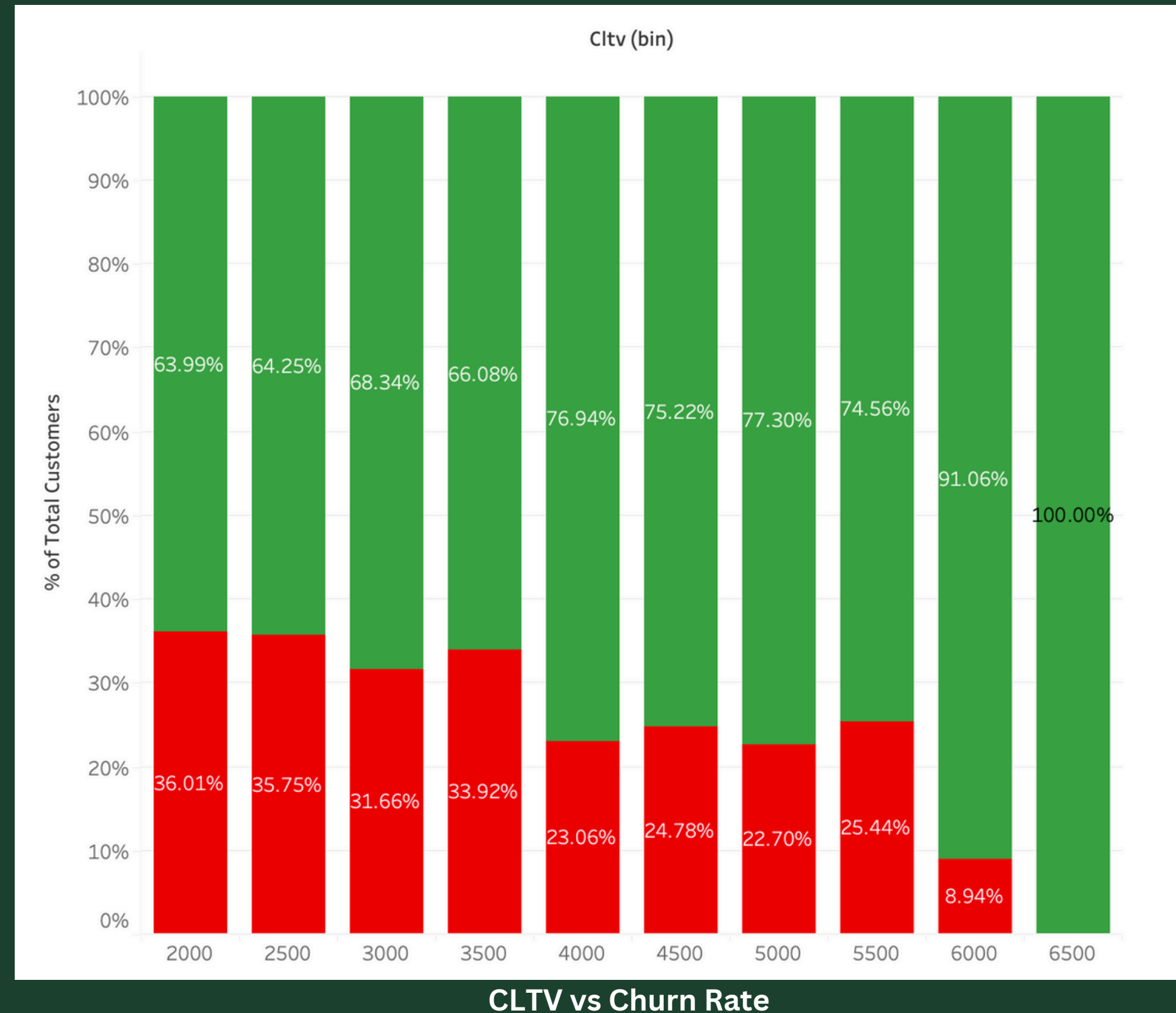
Internet Type vs Churn Rate

CLTV vs Churn

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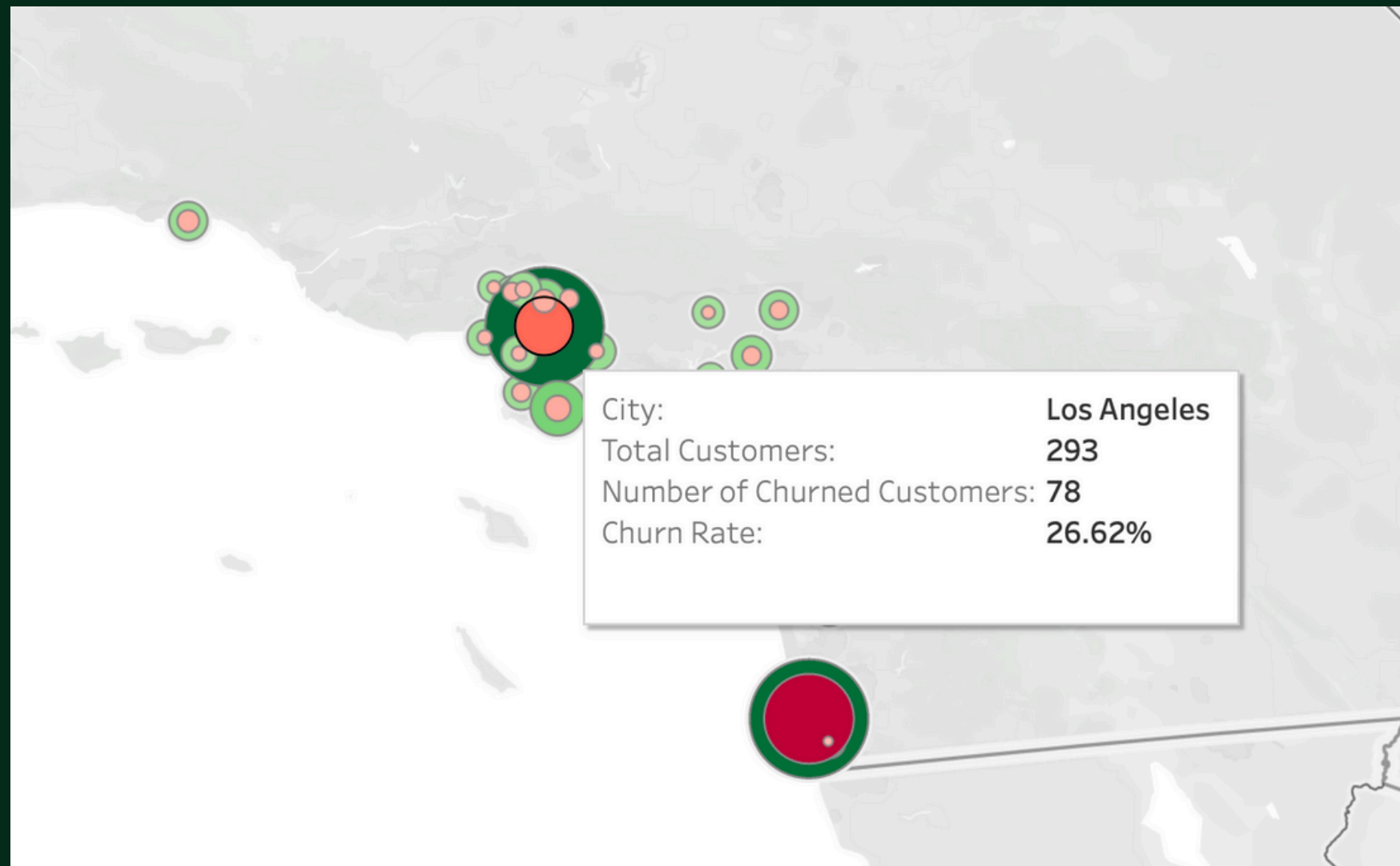
Customers with **lower CLTV (≤ 3500)** have **higher churn rate** than the **above 3500 CLTV**

- Customers with **lower lifetime value** are **more likely to churn** when compared to high value customers (> 3500)

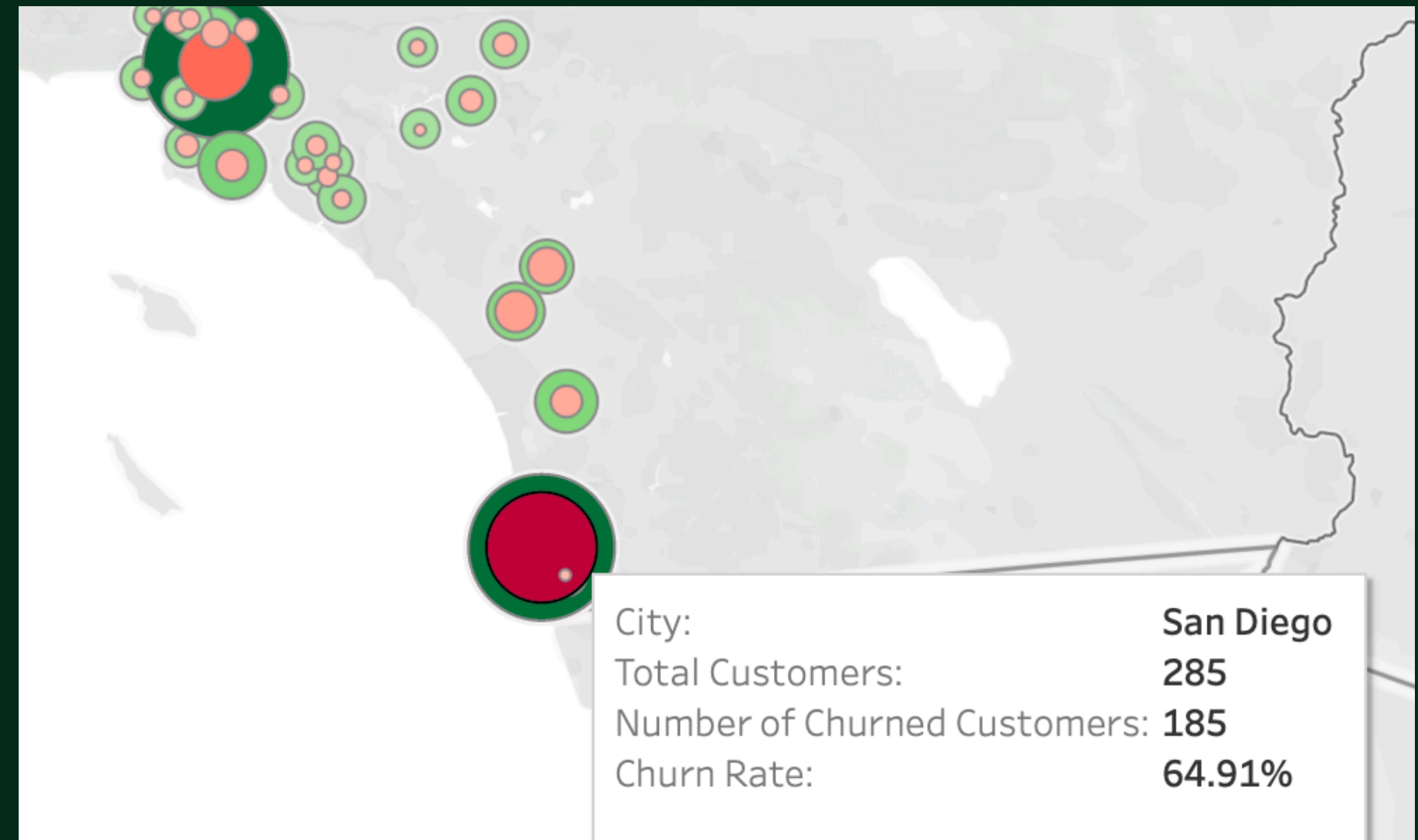


Cities vs Customer Churn

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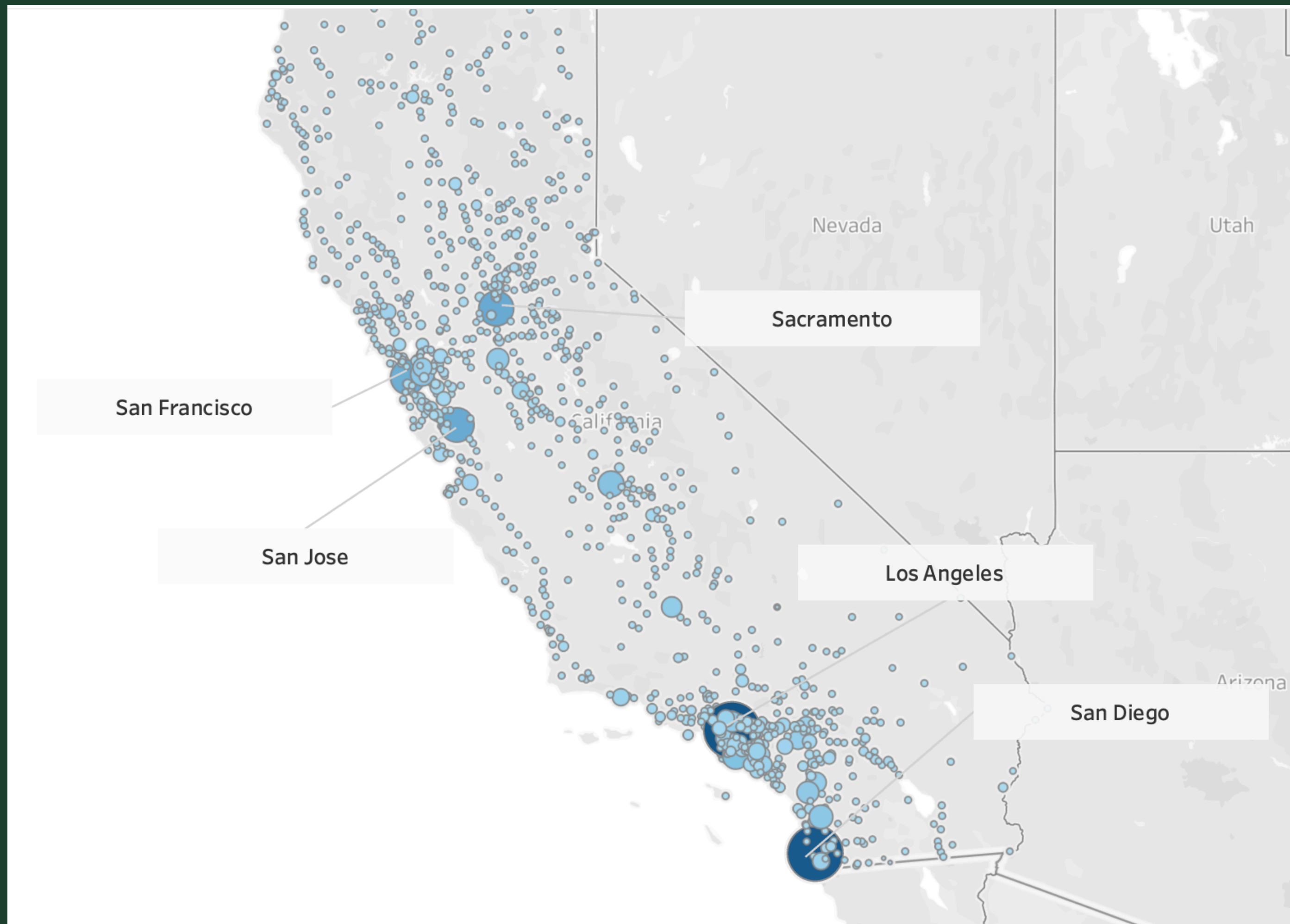


Los Angeles



San Diego

- **San Diego** has the **highest churn rate (65%)**
- **Los Angeles** has the **highest number of customers, followed by San Diego.**
- All the other cities have a churn rate nearly around the overall churn rate.
- A competitor has entered these cities where the company has many customers, and already started to offer better devices and offers, which are a major reason for customers to churn in these cities.



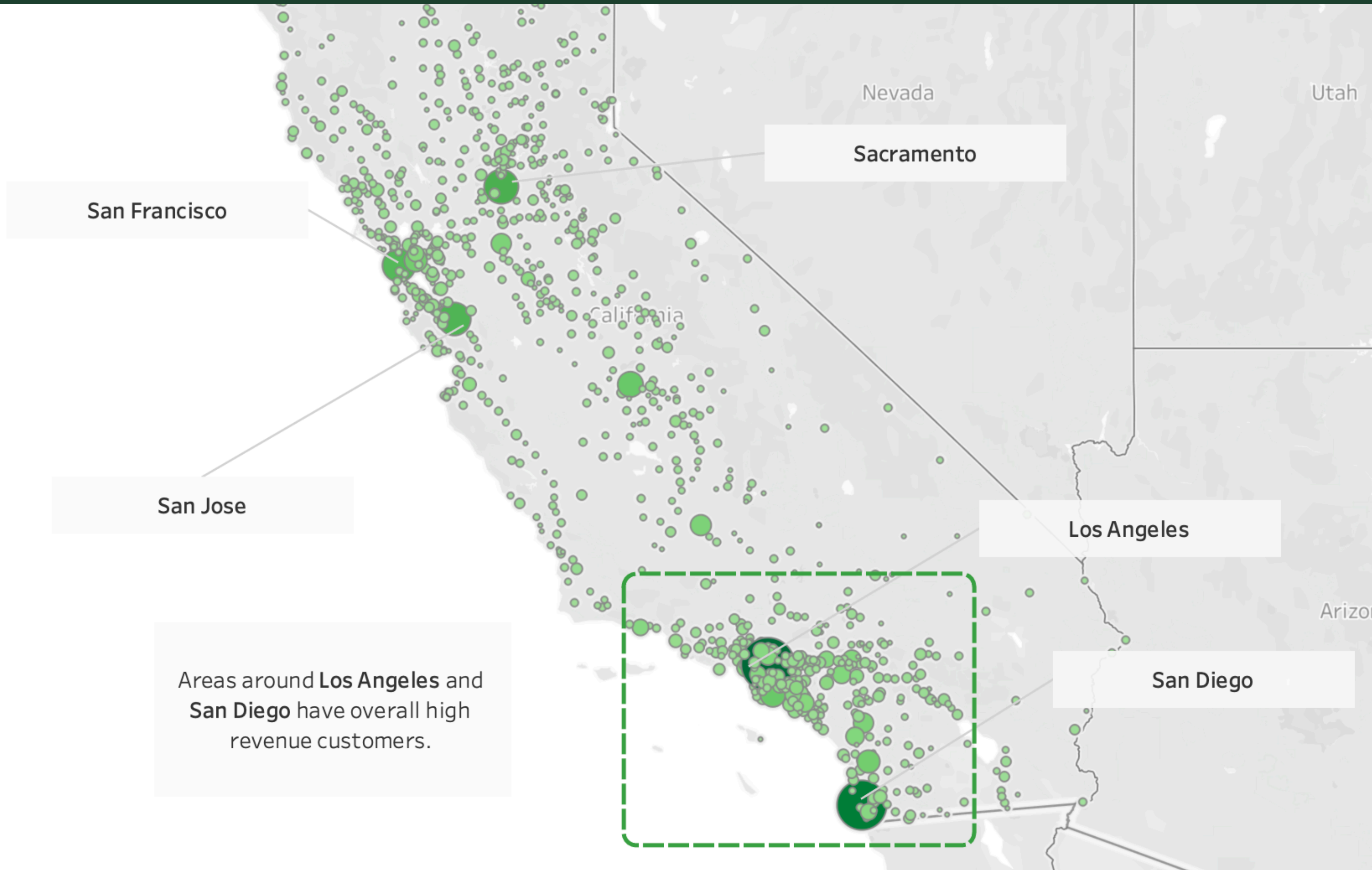
CLTV vs City

Insights?

Los Angeles and **San Diego** has highest overall CLTV (sum) in all cities

- **Los Angeles** followed **by San Diego, Sacramento, San Jose** and **San Francisco**, are the **cities** where the **highest overall customer value comes from**.
- Thus, if company decides to make changes and launch their retention campaign, these places would be ideal.





Revenue vs City

Insights?

Los Angeles and San Diego has **highest overall revenue** in all cities

- **Los Angeles** followed by **San Diego, Sacramento, San Jose** and **San Francisco**, are the cities where **the highest overall revenue** comes **from**.
- Also, **areas around LA and San Diego** are **high revenue areas**, so campaigning must be done in these areas first.





Better Offers and Devices

Conduct market research to understand competitors' offerings and devices. Introduce competitive pricing, better data plans, and technologically advanced devices to match or surpass competitors' offerings.



Attitude Issues

Invest in training programs for customer support representatives to improve their attitude and interaction quality with customers. Implement regular feedback mechanisms to monitor and enhance service quality.



Senior-Friendly Services

Design and promote services that cater specifically to the needs of senior customers, such as simplified plans, senior discounts, and dedicated customer support for seniors.



Enhance Streaming Quality

Invest in improving the quality and reliability of streaming services for music and movies. Ensure high bandwidth and minimal buffering for a better customer experience.



Promotional Campaigns

Launch targeted promotional campaigns in high churn areas (e.g., San Diego) emphasizing the improved value and benefits of your services compared to competitors.



Encourage Longer Contracts

Provide incentives for customers to switch from month-to-month contracts to one-year or two-year contracts. Offer discounts, additional services, or exclusive deals to long-term contract customers.

Continued...

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Device Upgrades

Address the issues with current devices by investing in better quality modems and other hardware. Ensure that new devices meet high performance and reliability standards.



Expand Offer A

Given that customers with Offer A have the lowest churn rate, expand this offer to a broader customer base. Tailor similar attractive offers to different customer segments based on their usage patterns and preferences.



Address Fiber Optic Issues

Since a significant portion of churned customers used fiber optic internet, investigate and resolve any underlying issues with fiber optic services. Ensure consistent high-speed internet and minimal service interruptions.



Tailored Marketing

Create marketing campaigns that appeal specifically to customers without dependents, highlighting features and benefits that resonate with their lifestyle and needs.



Local Marketing Efforts

Concentrate marketing efforts and resources on high-value cities like Los Angeles, San Diego, Sacramento, San Jose, and San Francisco. This can include localized advertising, sponsorships, and community events.

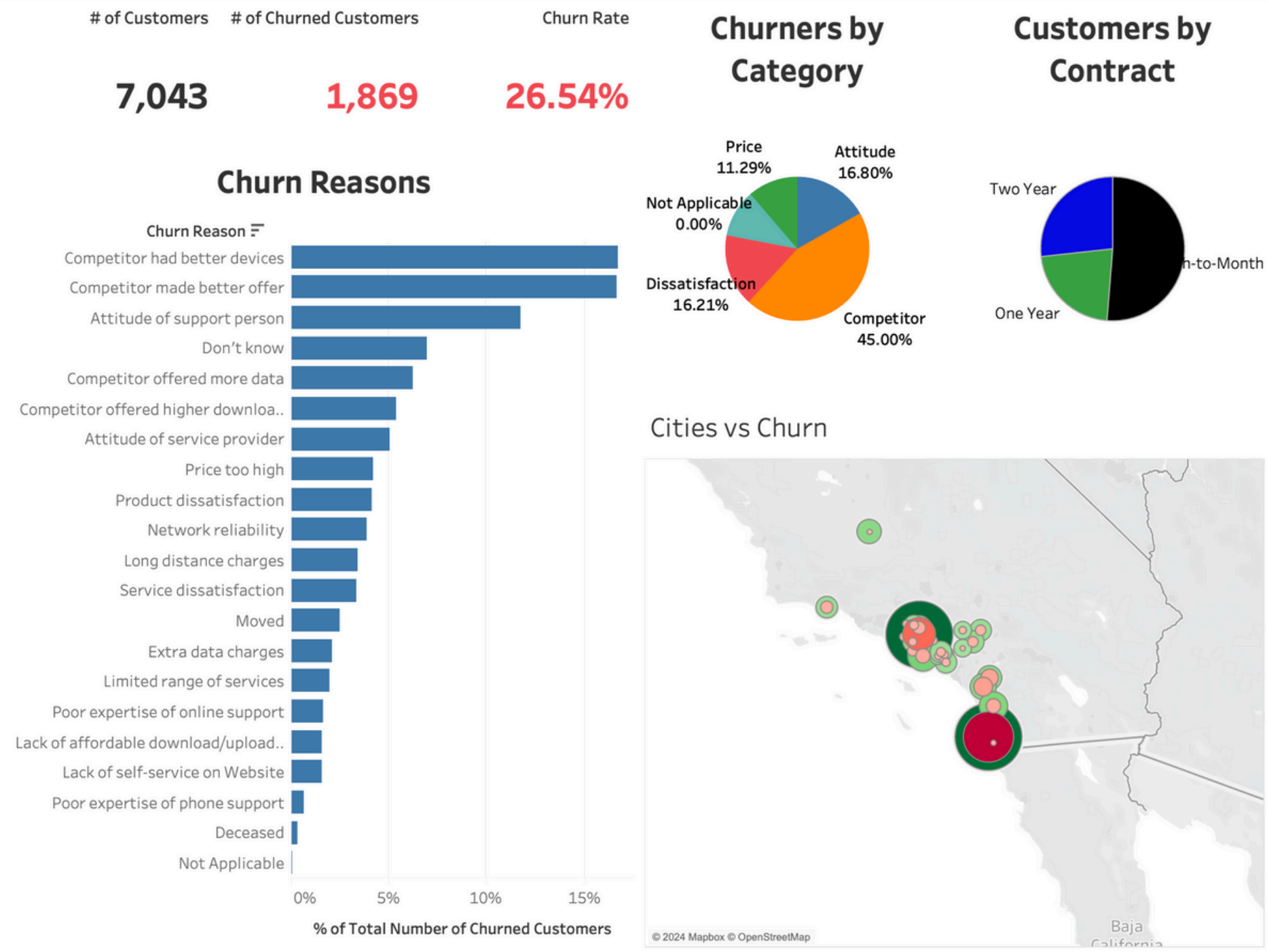


Contract Offers

Given the lower churn rate among one-year and two-year contract customers, create attractive offers for month-to-month customers to switch to longer-term contracts. This can include discounted rates, additional features, and loyalty rewards.

Dashboard Snaps

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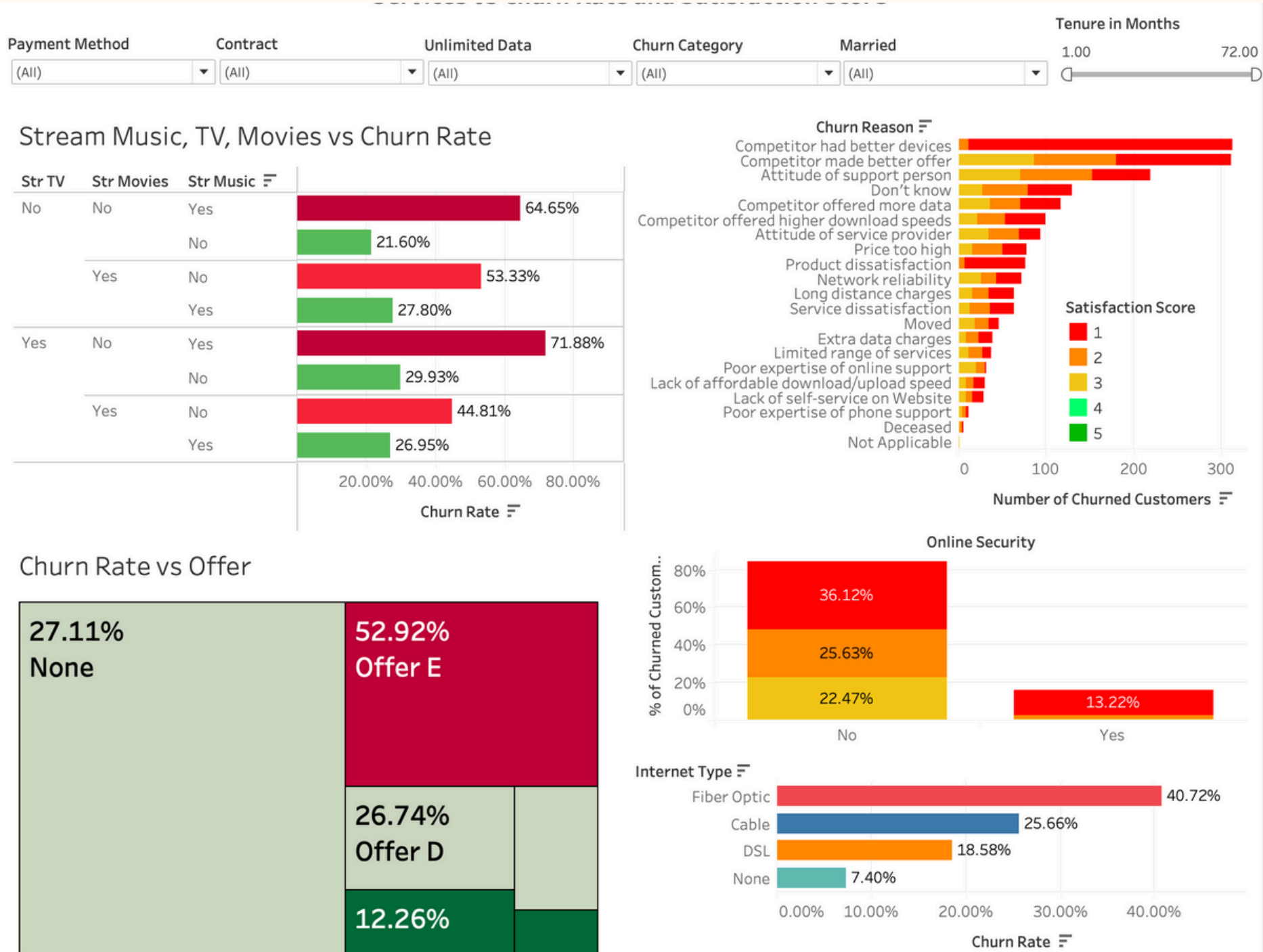


Overview

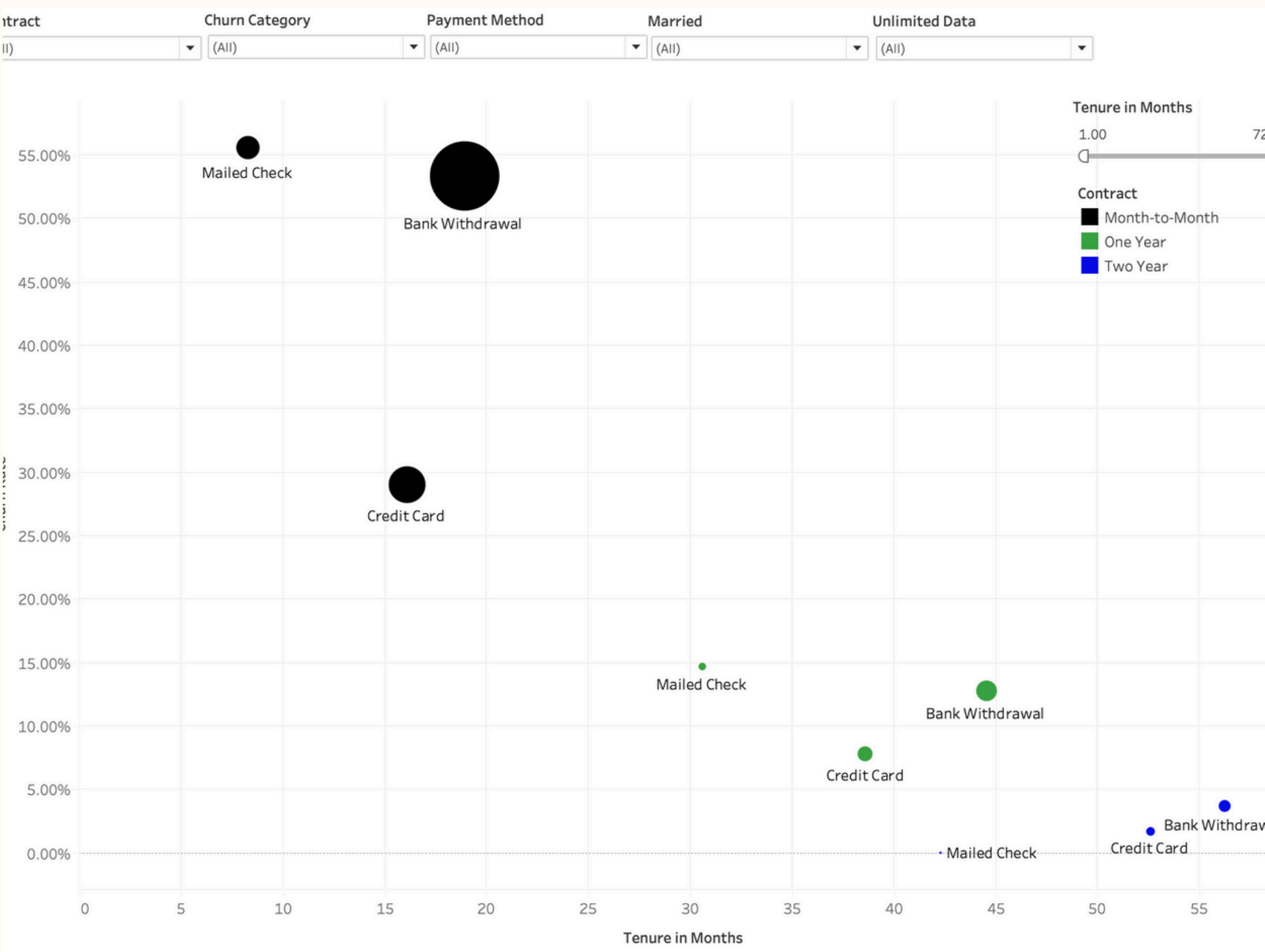
Age Brackets

Continued...

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Services vs Churn



Tenure in Months

1.0072

Contract

Month-to-Month

One Year

Two Year

Churn Rate

Tenure in Months

Mailed Check

Bank Withdrawal

Credit Card

Mailed Check

Bank Withdrawal

Credit Card

Mailed Check

Bank Withdrawal

Credit Card

Mailed Check

Bank Withdrawal

Credit Card

Payment Method & Contract Type

Thank
you

Get In Touch

Email

Rajatbahuguna4445@gmail.com

